



Scuola Superiore
Sant'Anna

Master of Science in Bionics Engineering

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Bioinspired and Soft Robotics

Soft Robotics Technologies

Shape Memory Alloys

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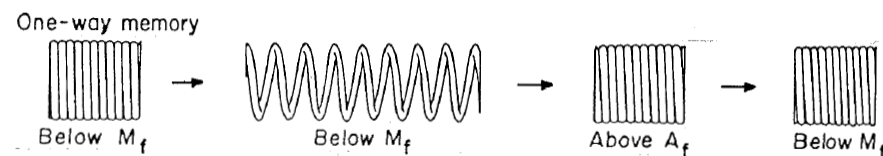


Shape Memory Effect and Superelasticity

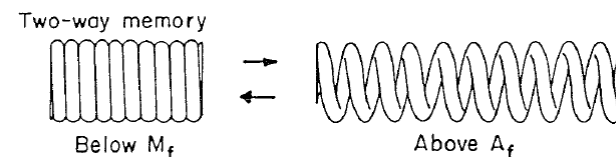
Shape Memory Effect (SME) refers to the recovery of shape (i.e. strain) after apparent “permanent” deformation (induced at relatively cold temperatures) by heating above a characteristic transformation temperature.

SuperElasticity (SE) refers to the isothermal recovery of relatively large strains during a mechanical load-unload cycle that occurs at temperatures above a characteristic transformation temperature.

One way (OWME)

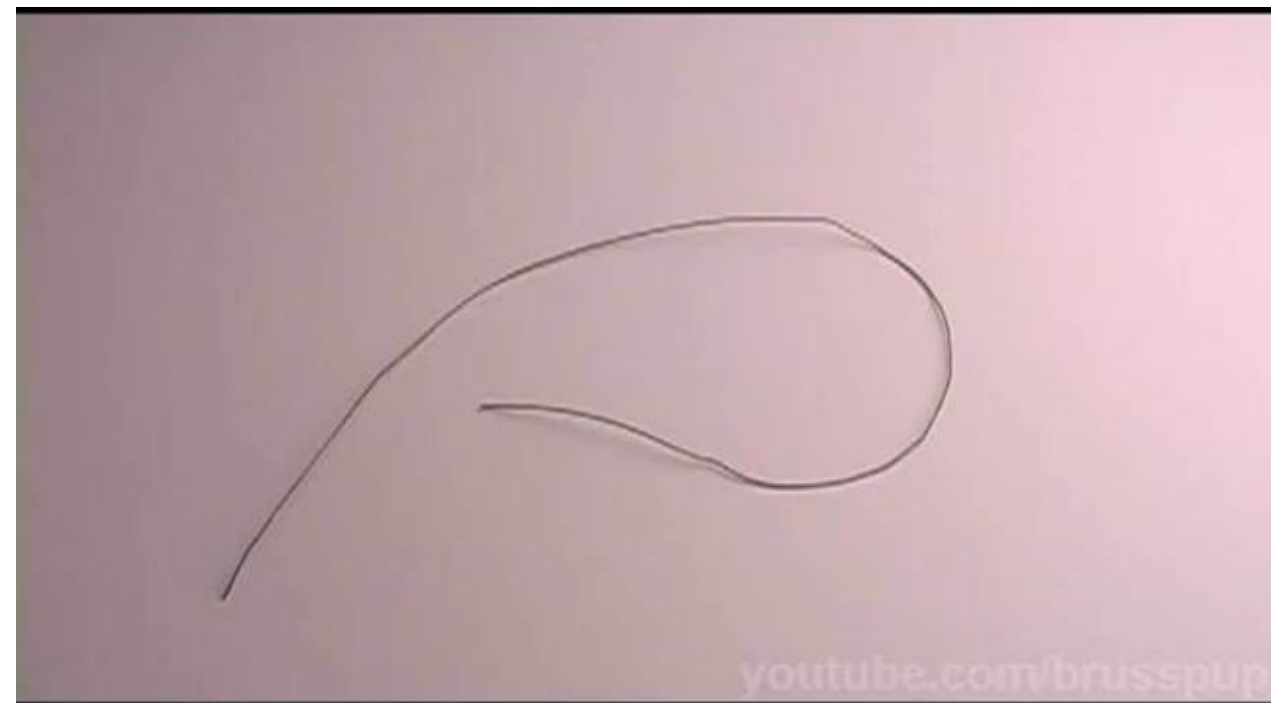
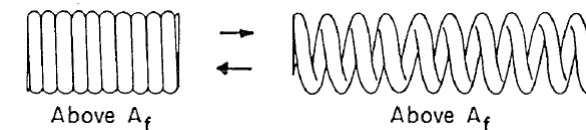


Two ways (TWME)



Note: this is an **acquired property**!

Training needed (thermo-mechanical treatments)



Brief history of Shape Memory Alloys

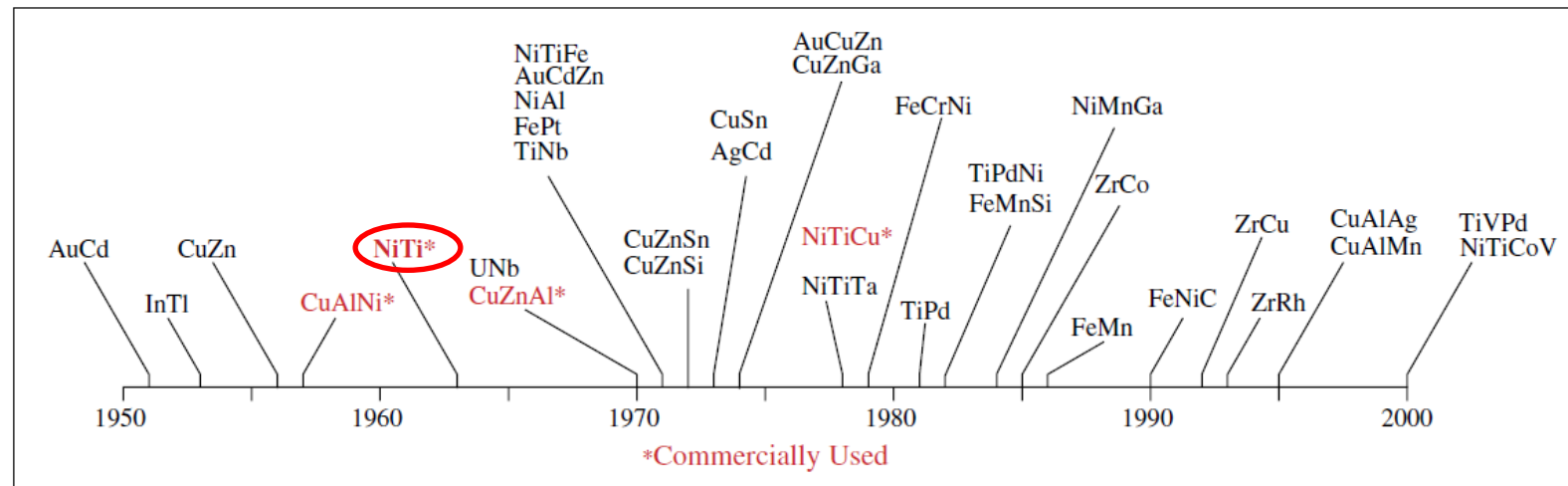
- 1932. Swedish physicist Arne Olander discovered the shape memory effect on an Au–Cd alloy
- 1958. Researchers Chang and Read demonstrated the shape memory effect at the Brussels World's Fair and used it to perform mechanical work by cyclically lifting a weight
- 1961. A group of U.S. Naval Ordnance Laboratory researchers led by William Beuhler discovered that an alloy of nickel and titanium exhibited the SME too. (Ni-Ti-NOL)
- 1970. First Commercial Use of NiTiNOL in F14 Tomcat



Shape Memory Alloys

Market

The field remains an active area, but advances in materials processing have resulted in production on Ni-Ti based materials with good quality control, with reproducible properties and in relatively large quantities



Shape Memory Alloys as actuators

PROs and CONs

PRO

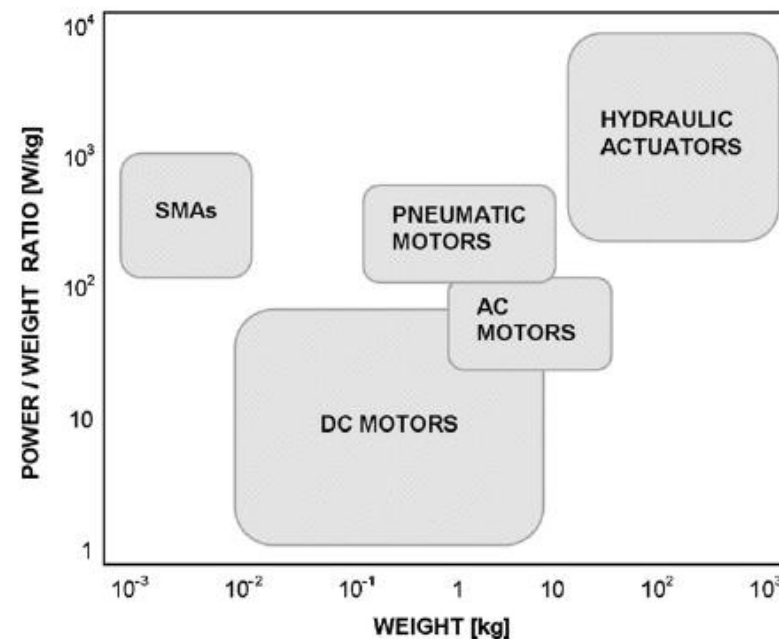
- Lightweight
- High work density
- Simplicity of mechanism
- Cleanliness
- Silent
- Remotability (distributed)
- Sensing capabilities
- Miniaturization
- Positive scale effects
 - response speed
 - power/weight ratio
- Low driving voltage
- Corrosion resistance
- Biocompatibility

PRO/CON

- Fabrication process
- Versatile
- Low cost

CON

- Low energy efficiency (1-10%)
- Fatigue (depending on operation range of strain)
- Low strain (5-8%, OWME – 2%, TWME)
- Relatively high current
- Low bandwidth
- Control issues
 - Nonlinear behaviour
 - Wide hysteresis

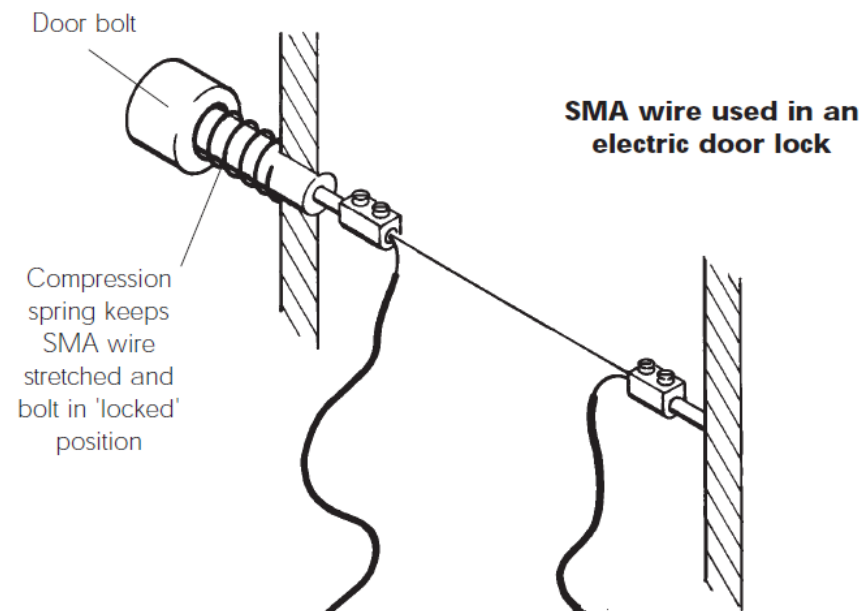


Depending on the application the cons can be limited

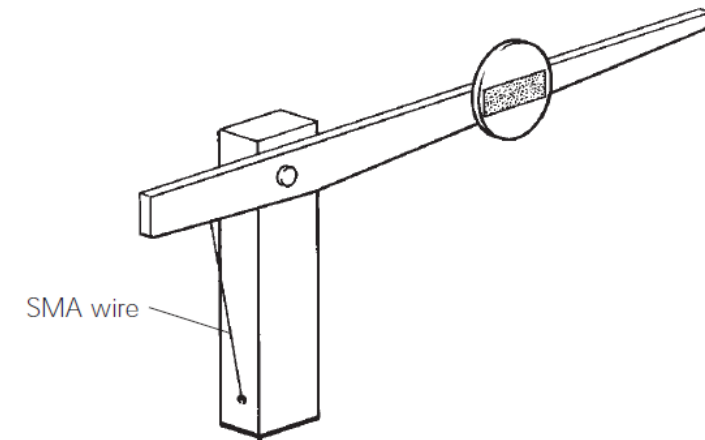


Applications

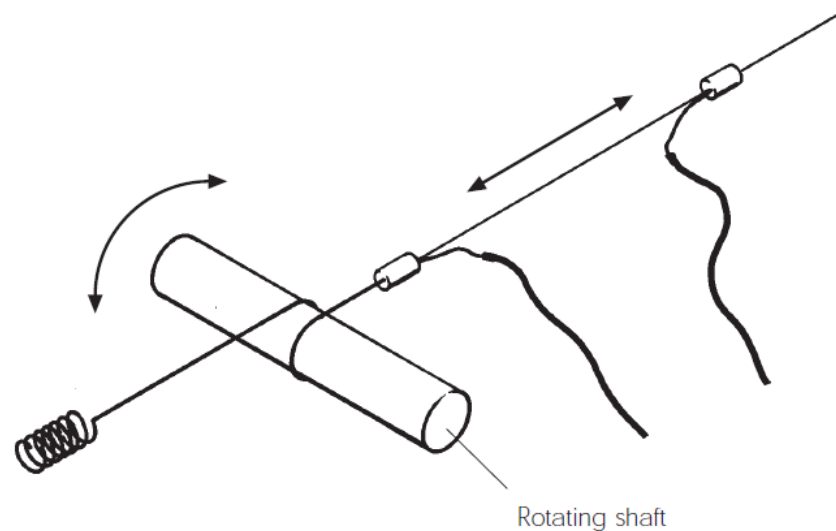
General



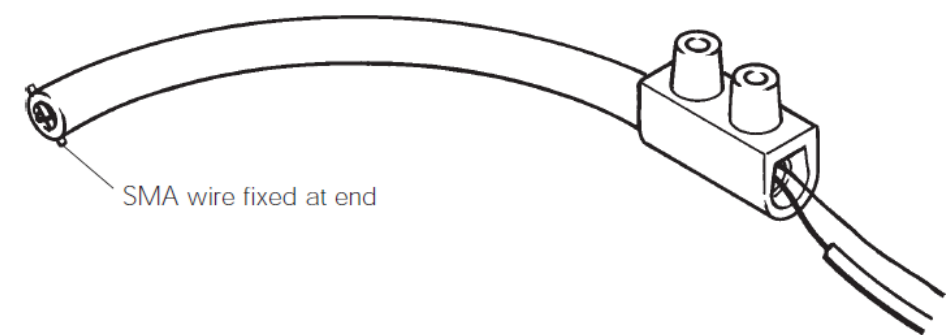
Linear actuation



Angular actuation



Rotary actuation



Bending actuation



Shape Memory Alloys actuators design

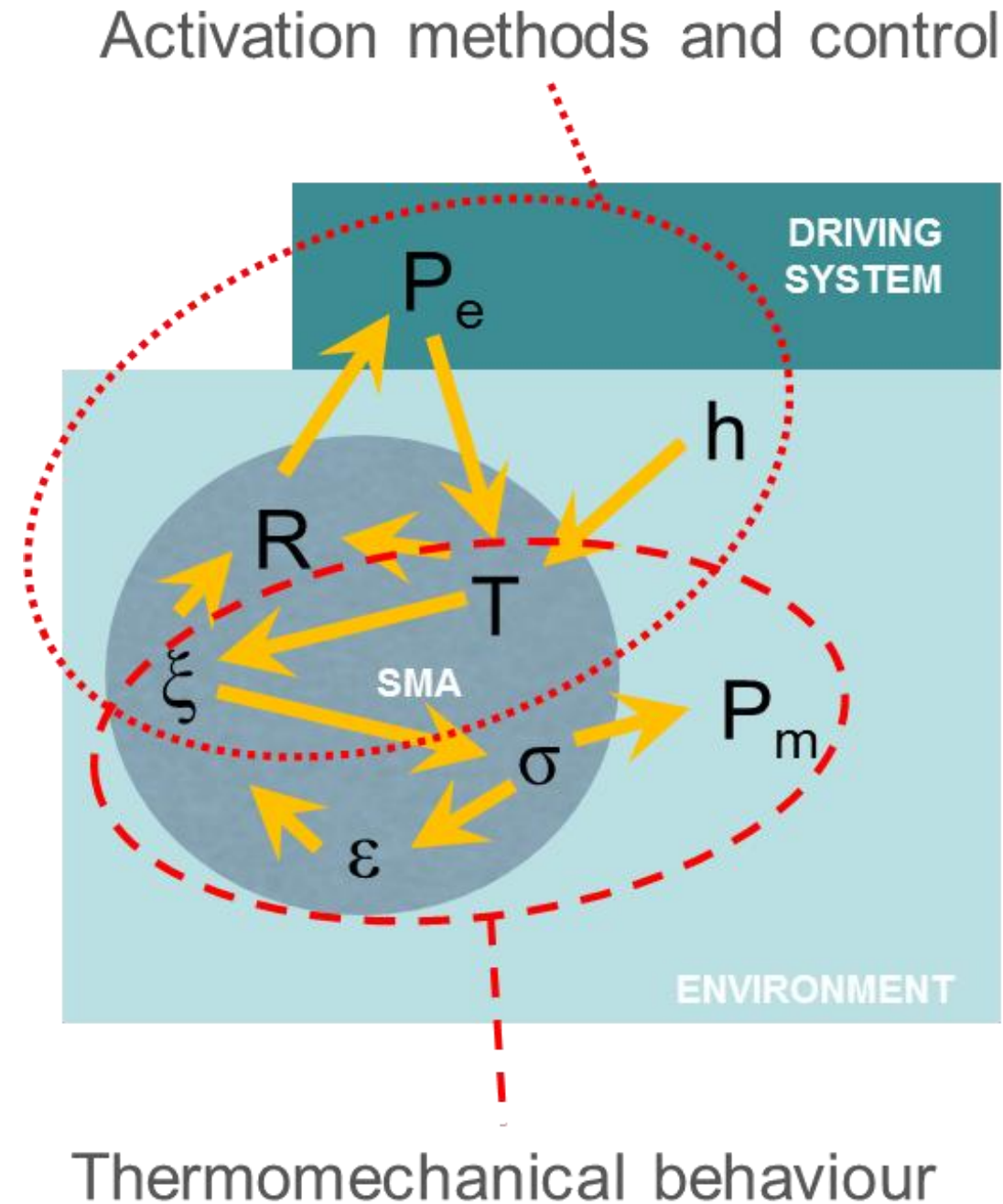
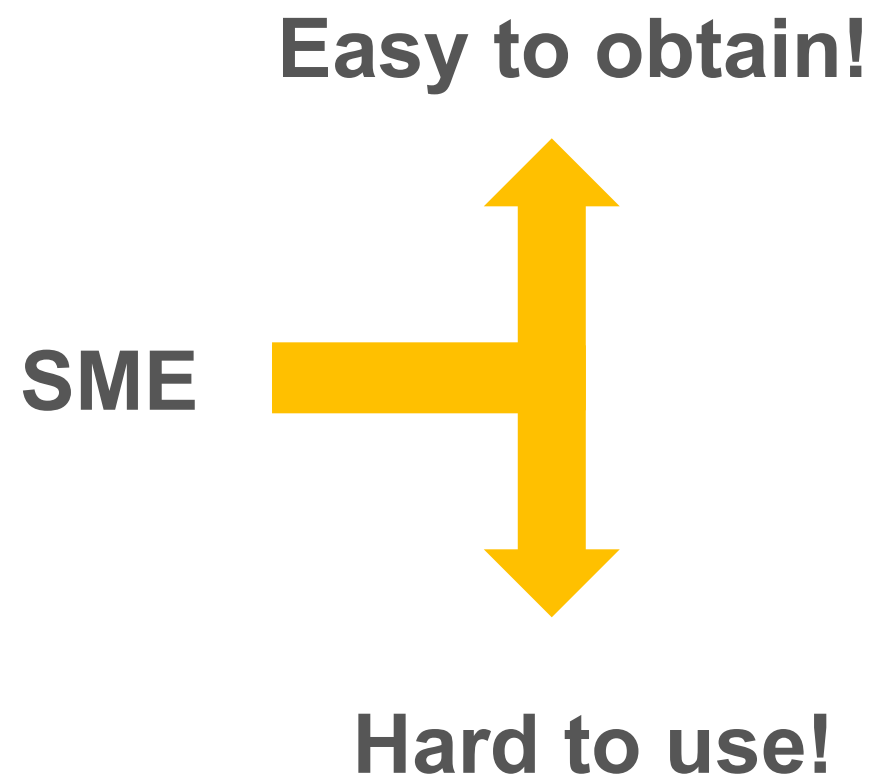
Workflow

1. Derive your actuator specifications
 2. Design the actuator (and the counter-actuator)
 3. Know your alloy
 4. Describe the environment
 5. Decide the activation and the cooling method
 6. Design the driving system
- MODELLING THE MECHANICAL PERFORMANCES OF THE ACTUATOR**
- MODELLING THE THERMODYNAMIC CYCLES DURING HEATING AND COOLING**
- DIMENSIONING THE CONTROL CURRENT**



Shape Memory Alloys actuators design

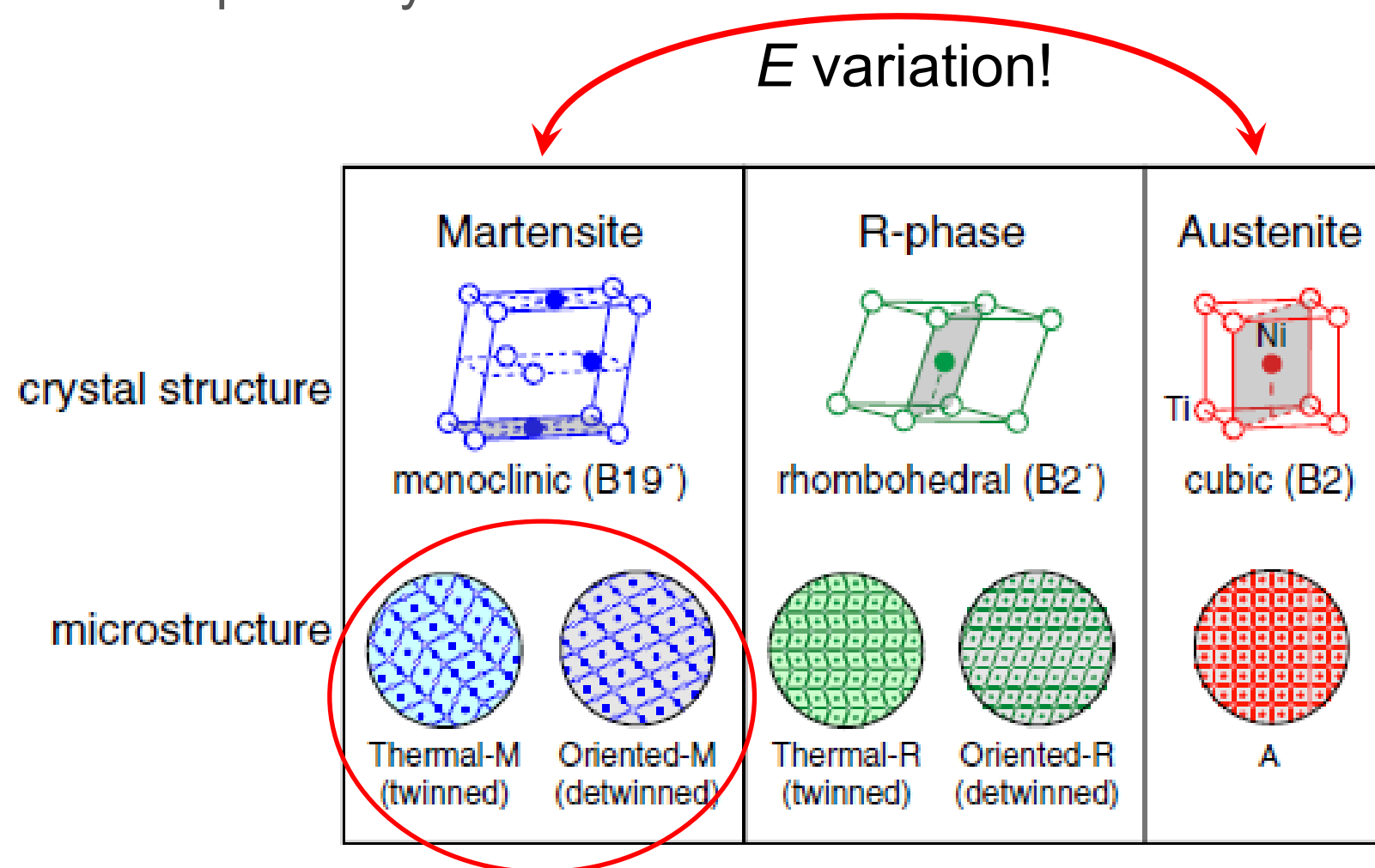
SME dependent variables flow chart



Shape Memory Alloys structure

Thermally activated martensitic transformation, diffusion less and reversible

The structure change can occur by small, coordinated shifts of the atomic positions without plasticity



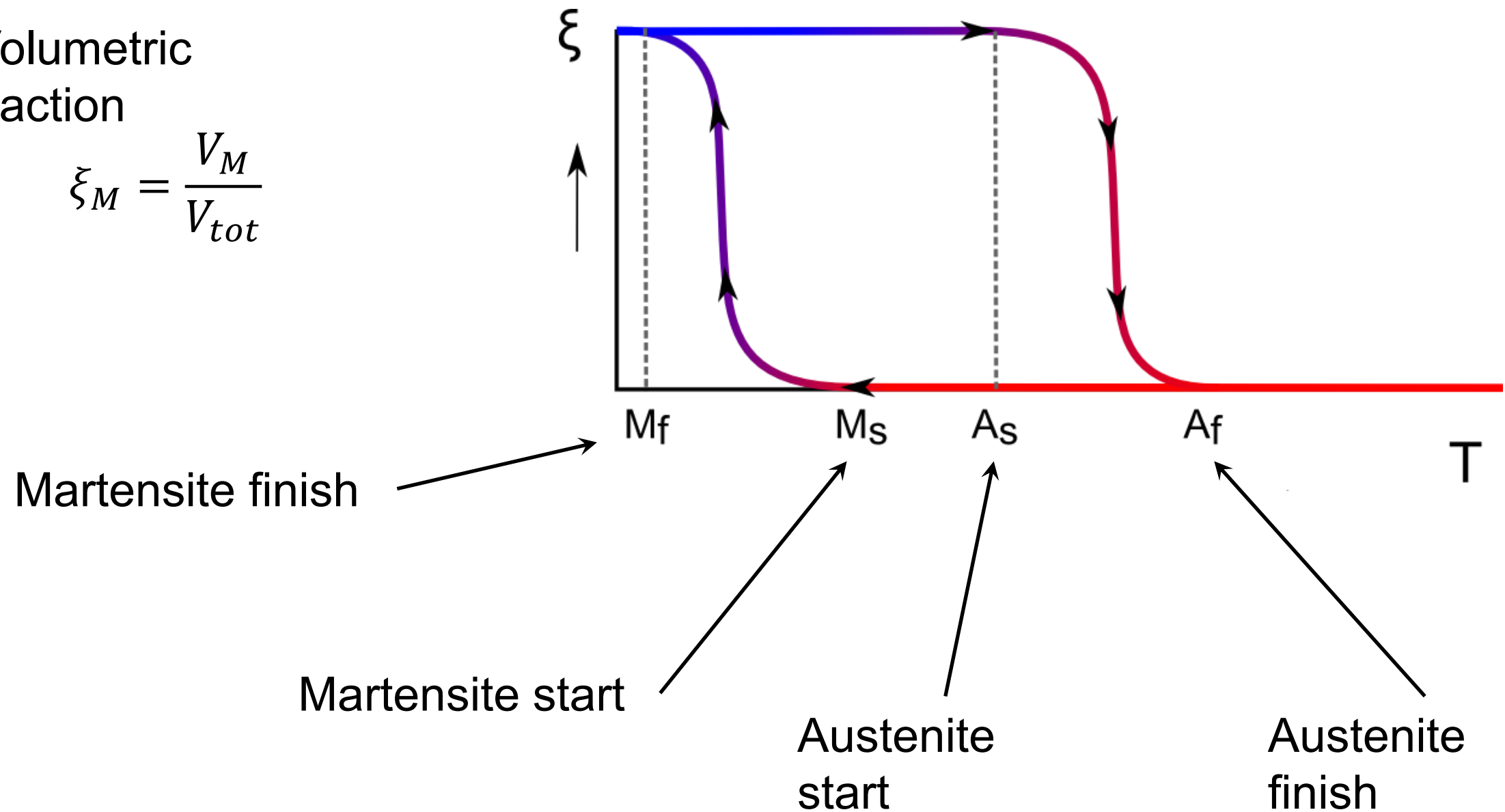
Large deformation!



Transition temperatures

Volumetric
fraction

$$\xi_M = \frac{V_M}{V_{tot}}$$

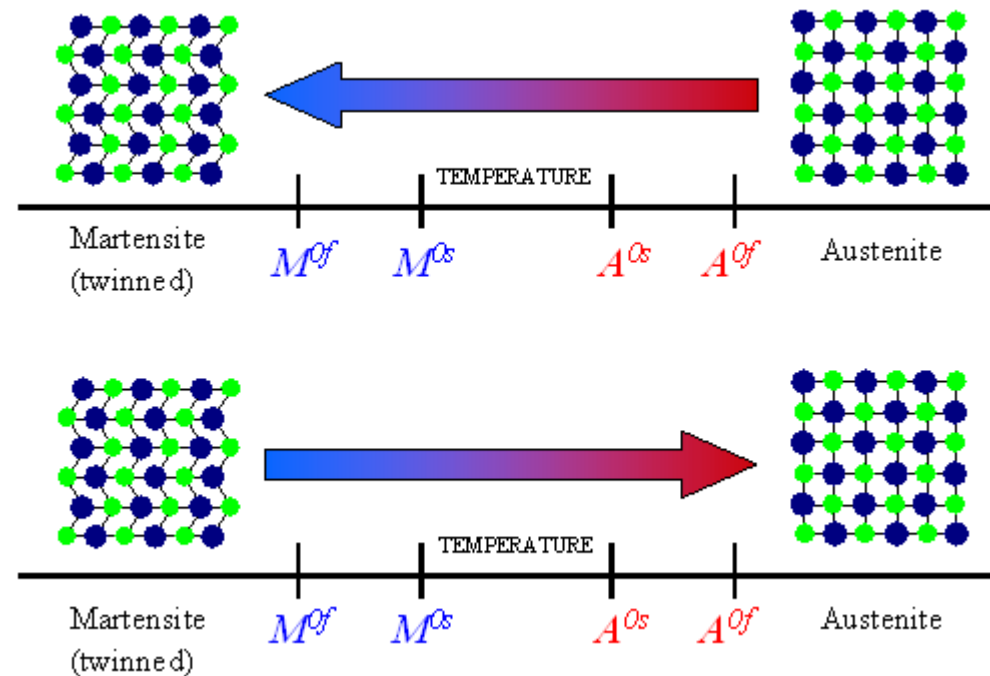


N.B. Several ranges and combinations available



Thermomechanical behaviour

Shape Memory Effect – without mechanical loading

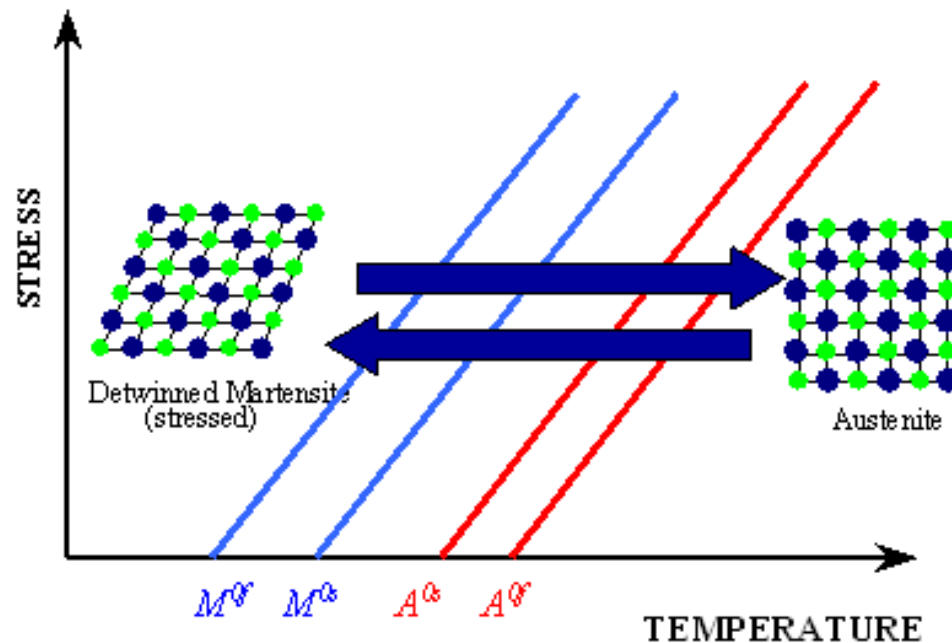


Temperature-induced phase transformation of an SMA without mechanical loading.



Thermomechanical behaviour

Shape Memory Effect – with mechanical loading (I)



Dependence of the transformation temperature on the applied load

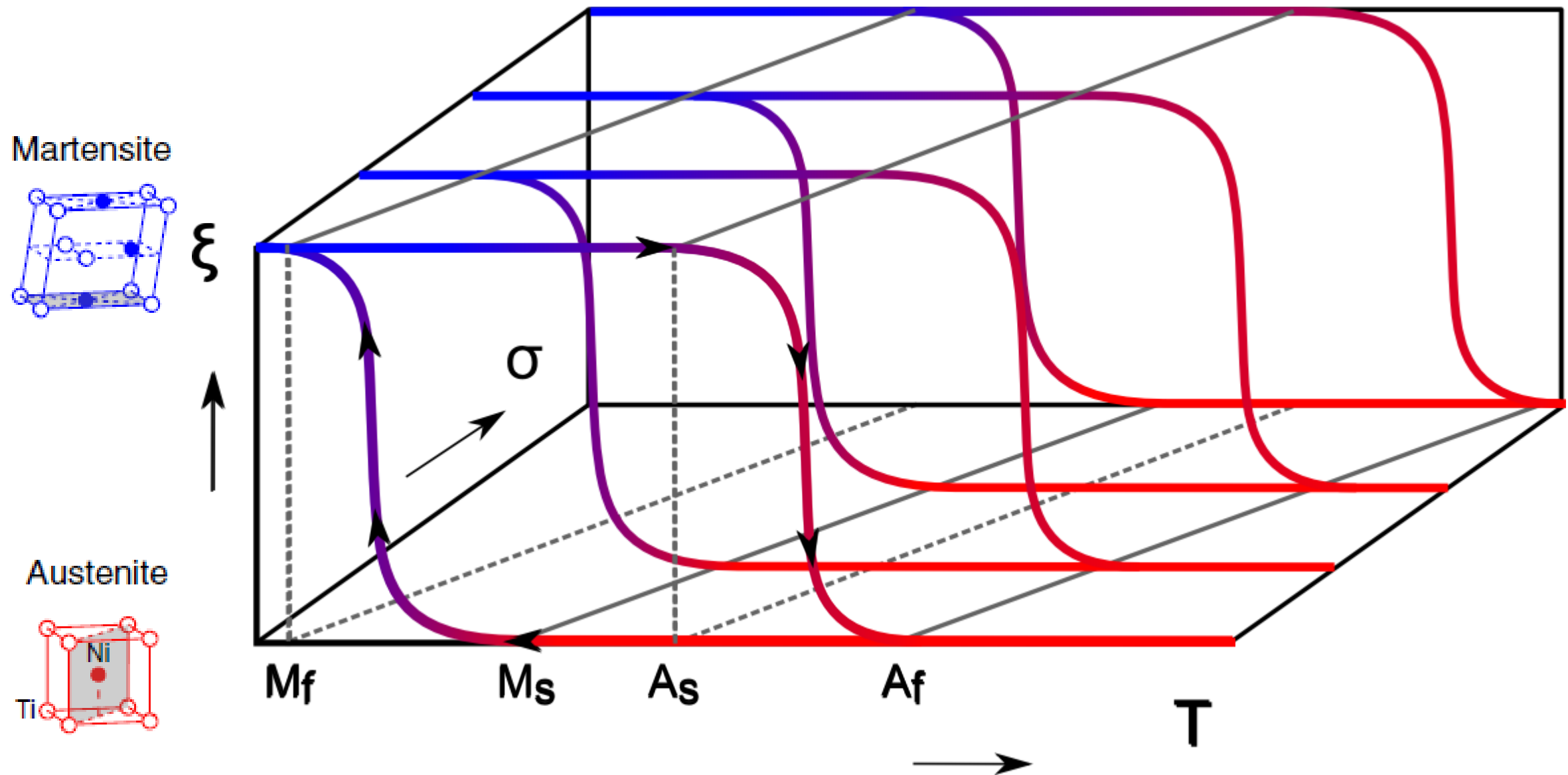
$$\frac{d\sigma}{dT} = \frac{\rho\Delta H}{T_0\Delta\varepsilon} = \frac{1}{c_m}$$

- ΔH : transformation latent heat
- T_0 : null stress transformation temperature
- $\Delta\varepsilon$: strain due to phase transformation
- c_m : stress gradient



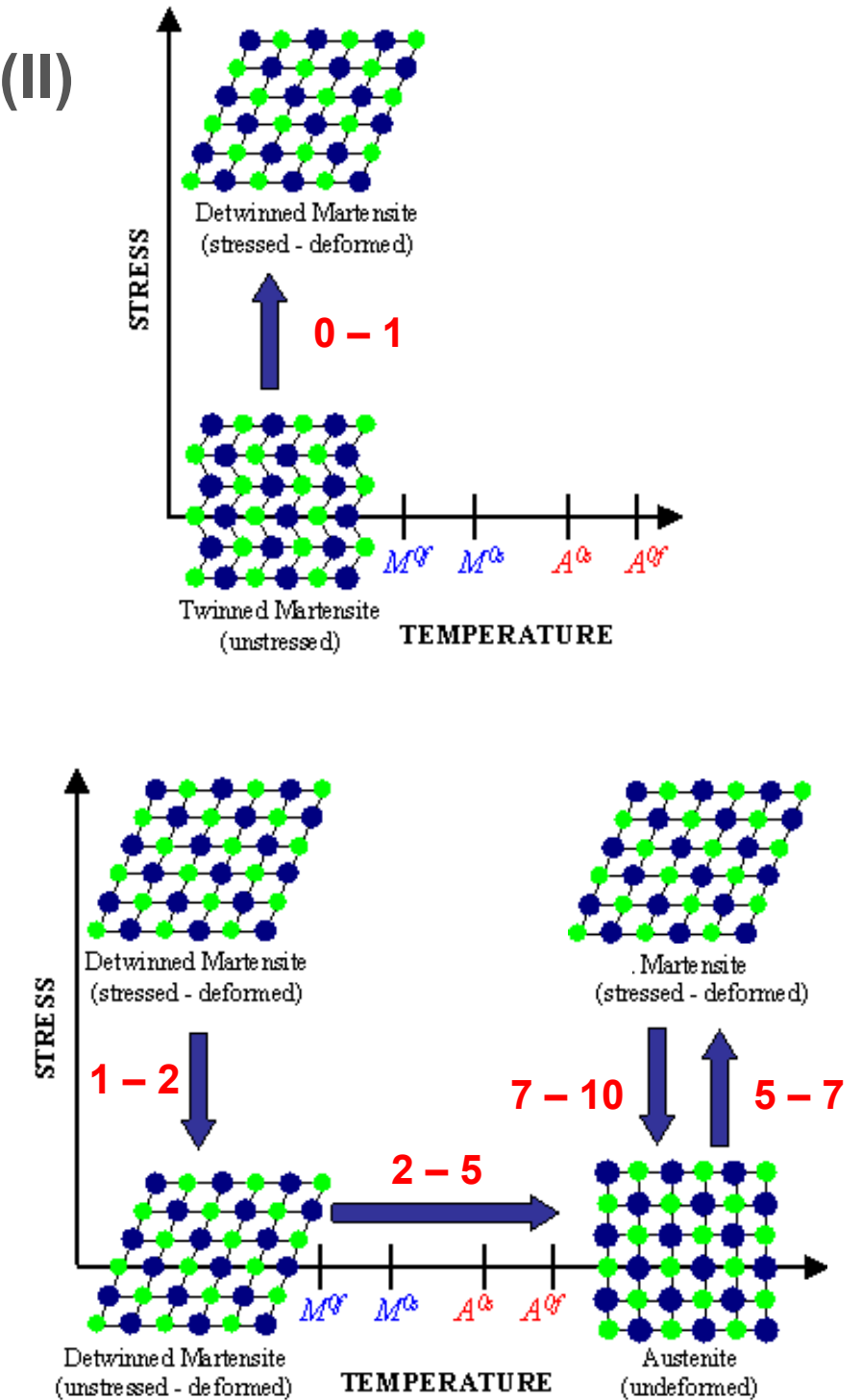
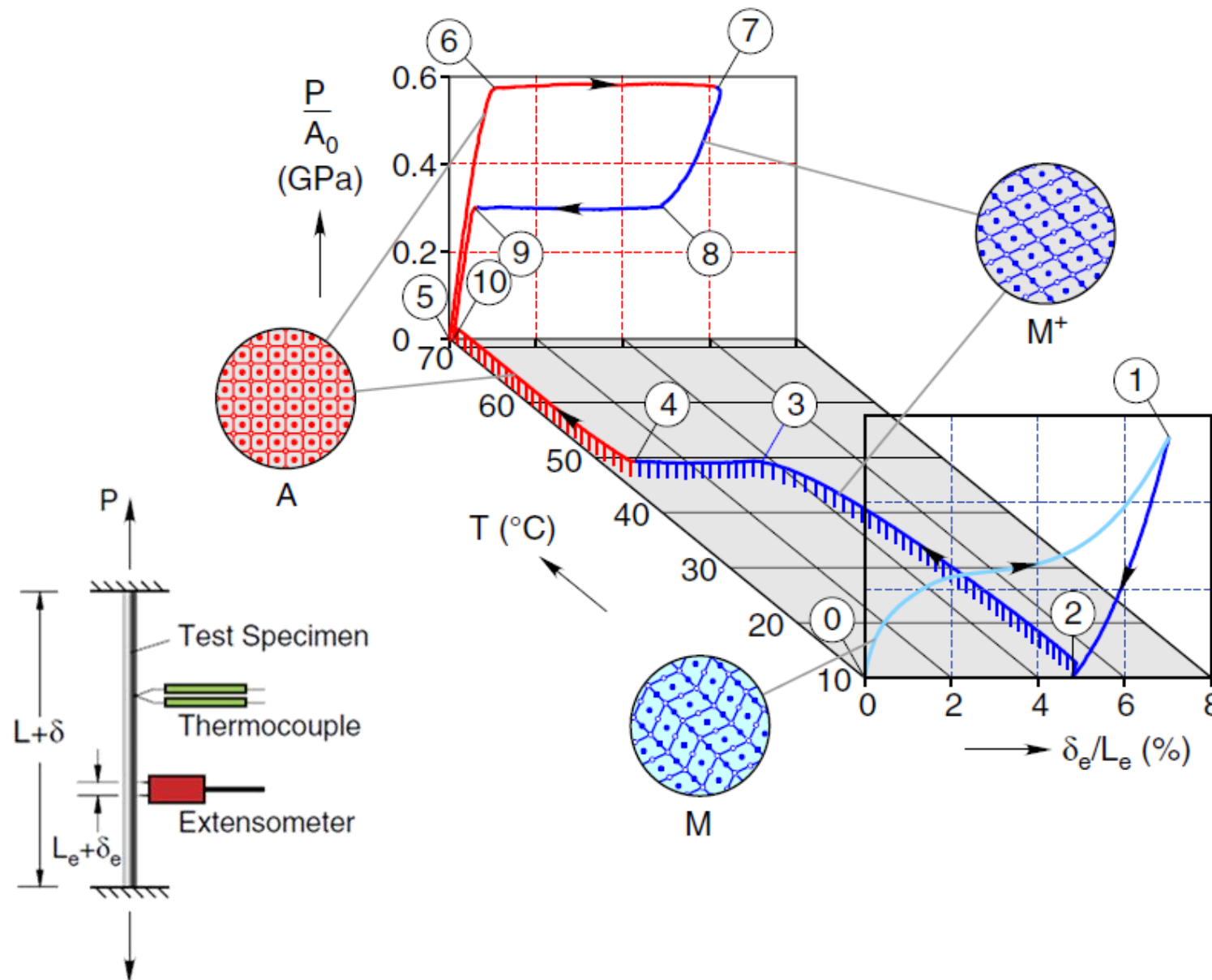
Shape Memory Alloys structure

Martensitic transformation cycles and “stress gradient”



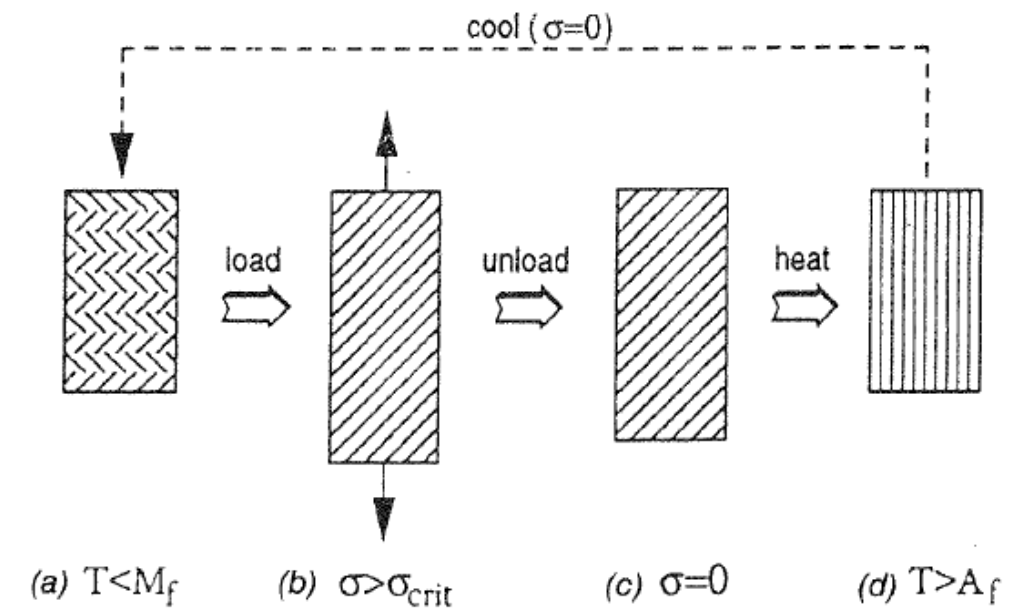
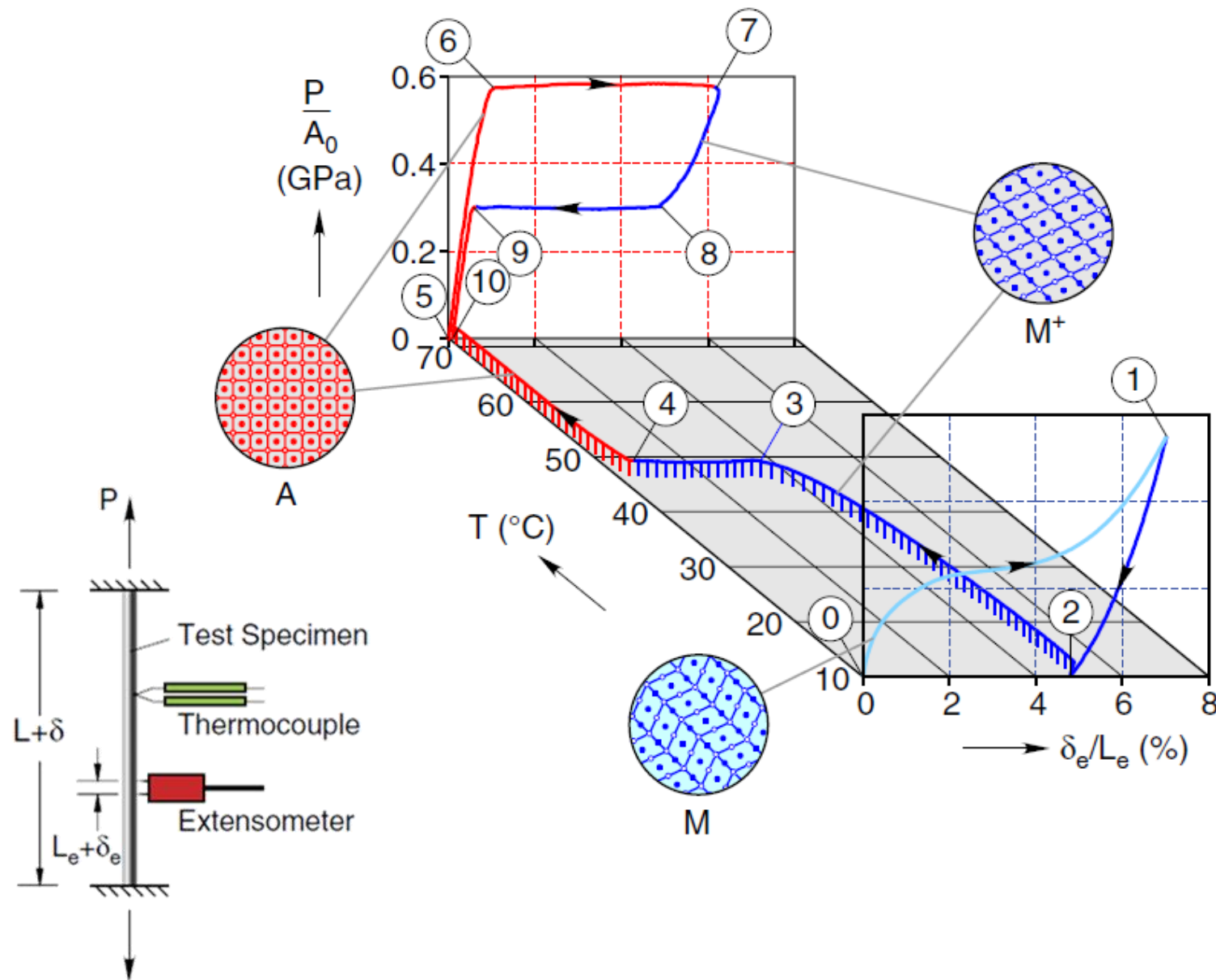
Thermomechanical behaviour

Shape Memory Effect – with mechanical loading (II)



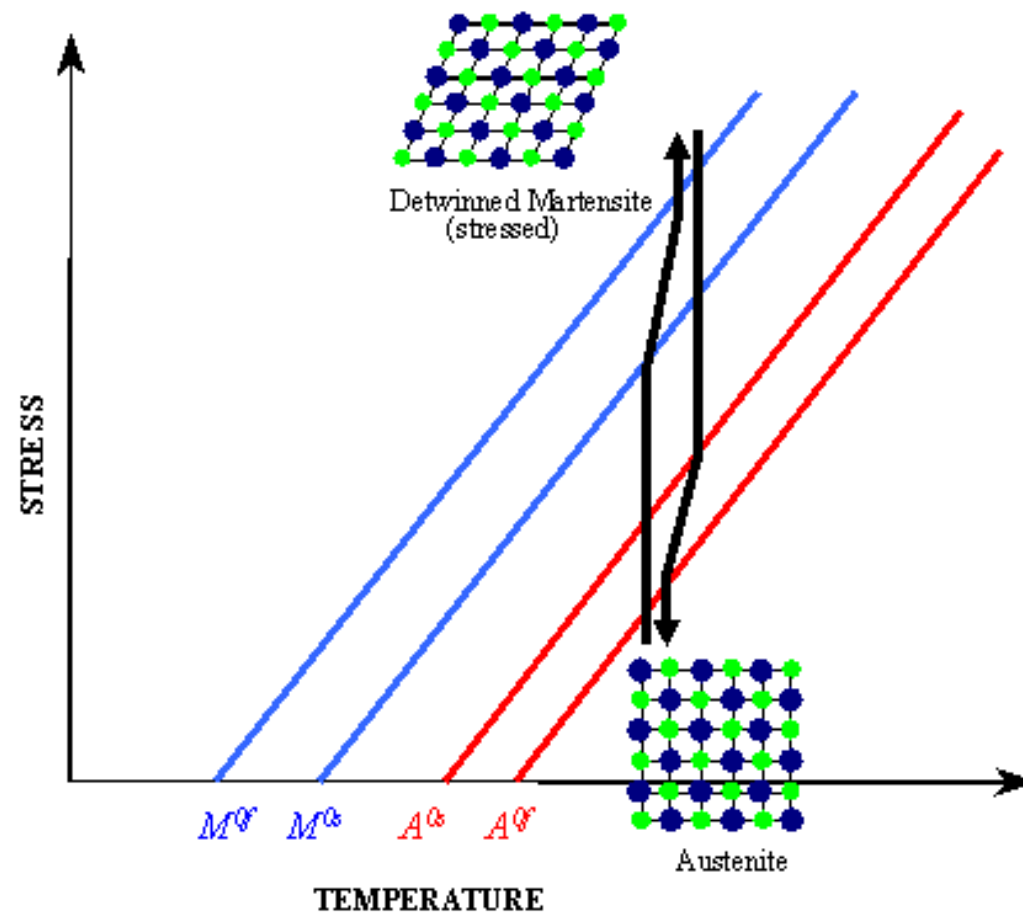
Thermomechanical behaviour

Shape Memory Effect – with mechanical loading (II)

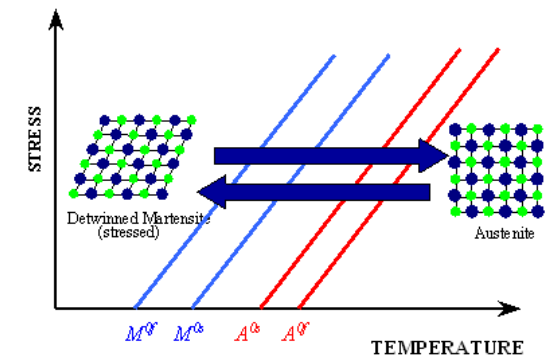


Thermomechanical behaviour

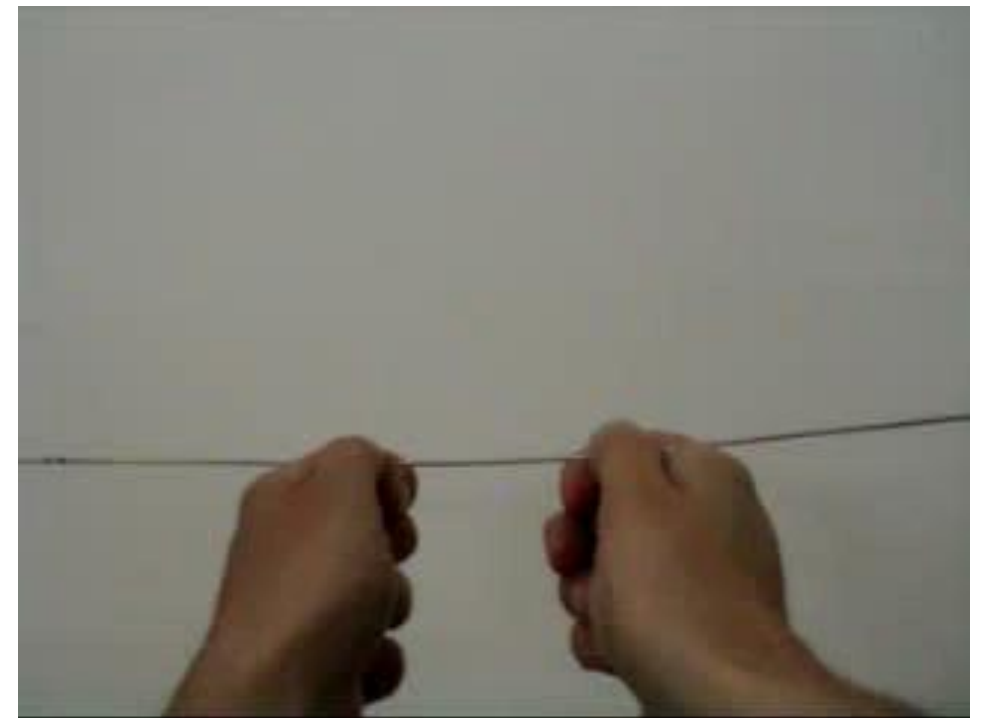
SuperElastic Effect (= Pseudoplastic effect)



Dependence of the transformation temperature on the applied load

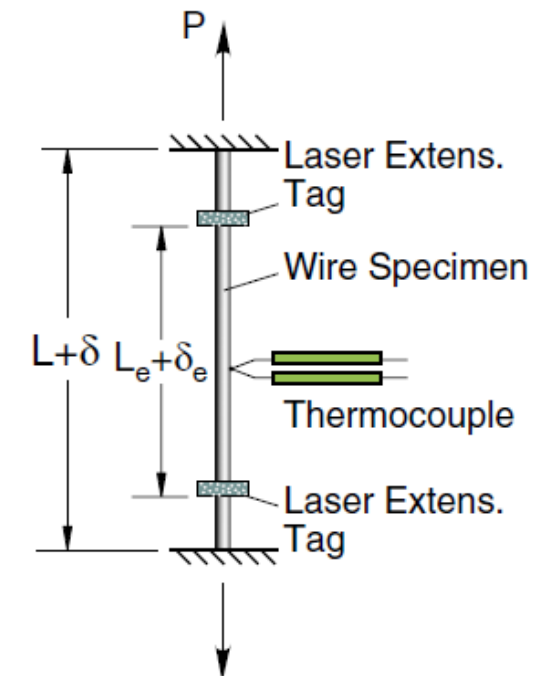
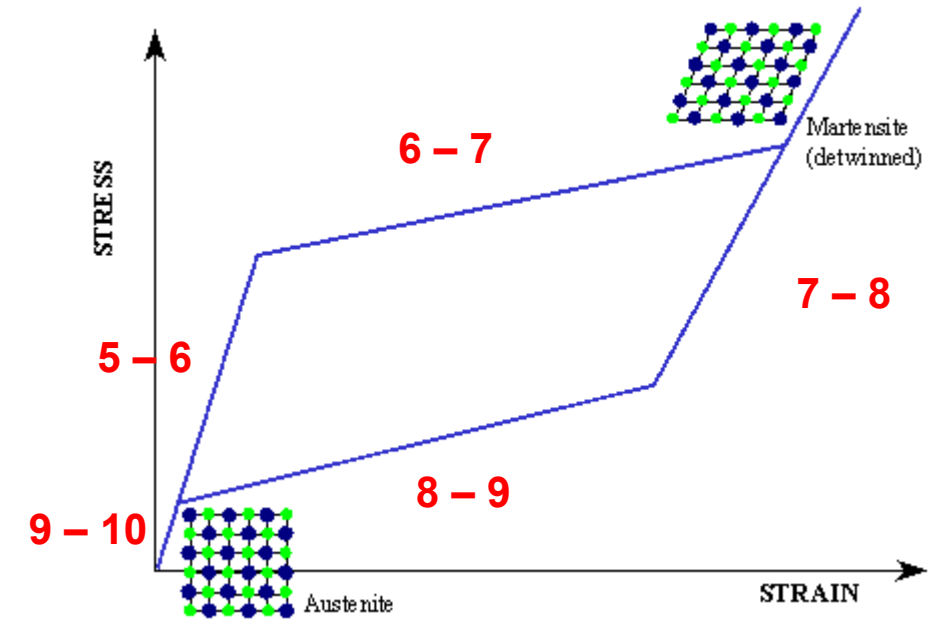
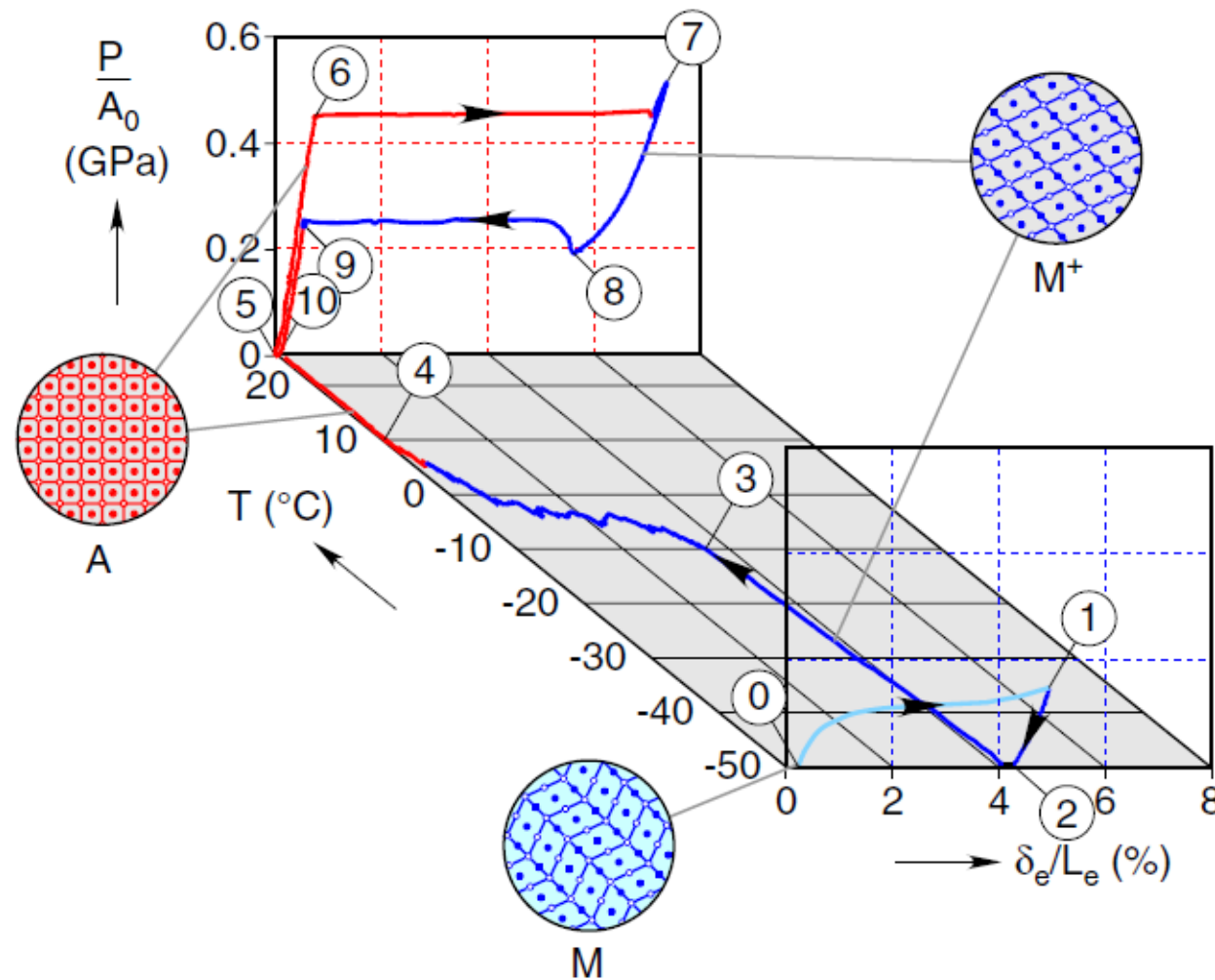


SME



Thermomechanical behaviour

SuperElastic Effect – with mechanical loading



Modelling the SME

- The SMA exhibits a **nonlinear** and **pseudo-elastic response** that varies depending on the temperature and loading conditions.
- An **accurate model** of this behavior is a necessary prerequisite for the use of the SMA as actuator.
- Models representing the thermo-mechanical behaviour and have mathematical expression in a form that is **amenable to incorporation into other engineering tools** (finite element procedures or control analysis programs).
- Due to the complex interactions of the different parameters controlling the SMA behavior, these materials have been the subjects of **numerous constitutive models**.



Modelling the SME

Classification

1. Microscopic (molecular level) or mesoscopic (lattice particles) phenomena

These approaches have been studied by several authors including Warlimont et al. [1974], Perkins [1975], Nishiyama [1978], Achenbach, and Muller [1982], Sun and Hwang [1993], Fischer and Tanaka [1992], Comstock et al. [1996], Lu and Weng [1998], Levitas et al. [1998], Gall et al. [1999], Sittner and Novak [2000], Kloucek et al. [2004], Muller and Seelecke [2001]

2. Macroscopic approach (phenomenological features) – **THE SIMPLEST**

Falk [1983], Falk and Konopka [1990] and The Landau-Devonshire theory

3. Models with assumed phase transformation kinetics (preestablished simple mathematical functions to describe the phase transformation kinetics) – **THE MOST POPULAR**

Tanaka and Nagaki [1982] that motivated other researchers who present modified transformation kinetics laws as Liang and Rogers [2000], Brinson [1993], Ivshin and Pence [1994], Boyd and Lagoudas [1996]

4. Models based on elastoplasticity theory

Simo and Taylor [1986]

5. Models based on plasticity concepts

Silva [1995], Souza et al. [1998], Auricchio et al. [1997], Govindjee and Kasper [1997] and Leclercq et al. [1995]



Modelling the SME

FROM THE PERSPECTIVE OF A CONTROL ENGINEER, ANY MODEL THAT CAN CAPTURE THE SMA RESPONSE ADEQUATELY AND EFFICIENTLY (POSSIBLY IN REAL TIME) IS A “GOOD” MODEL, IRRESPECTIVE OF WHETHER THAT MODEL HAS BEEN BUILT ON PHYSICAL PRINCIPLES OR PURELY EMPIRICALLY



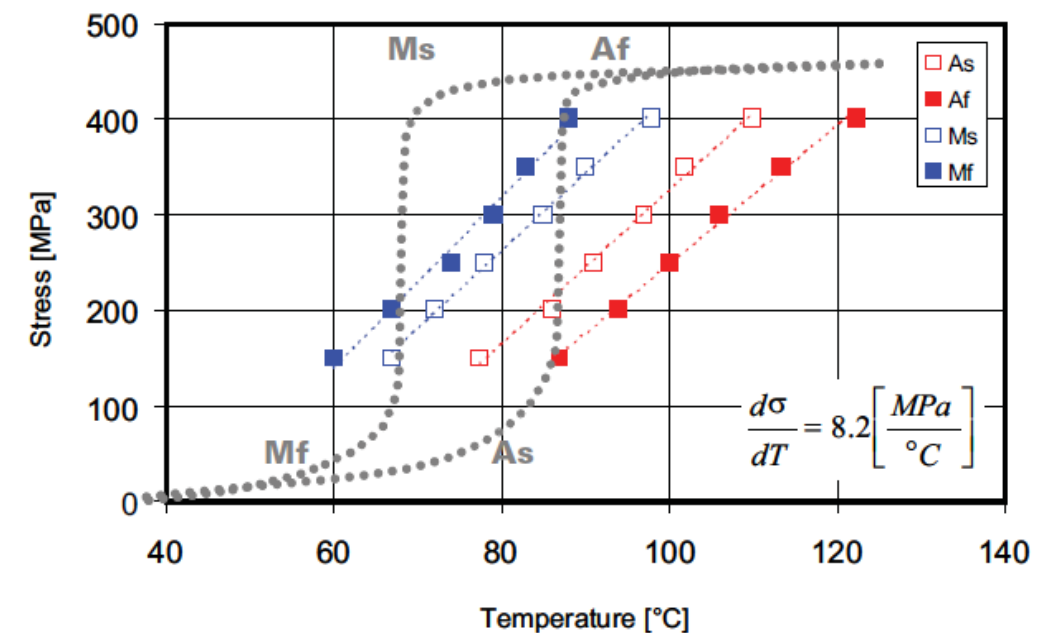
Modelling the SME

Useful data

1. Characteristic transformation temperatures
2. Young modulus of both phases (Martensite and Austenite)
3. Stress dependency of the transformation temperatures

The testing of SMAs is **not yet standardized** and unlike conventional alloys, material property tables either are not available or provide **incomplete**, or even **incorrect**, information for the user.

Since each SMA is different, user is often faced with **testing SMAs in their own laboratory** to obtain a satisfactory characterization of the material at hand.

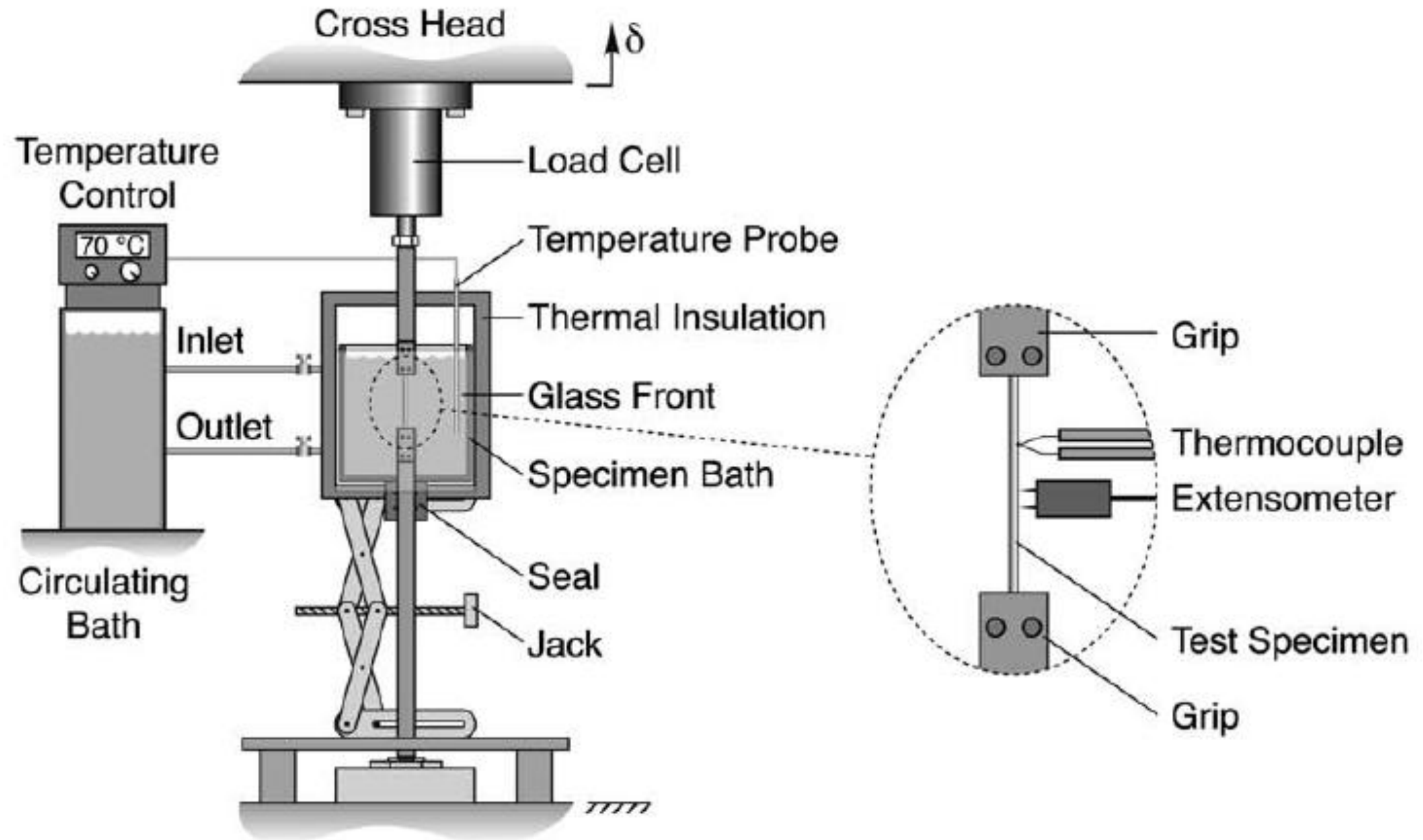


Transition T	-200 ÷ 100°C
Transformation strain	8% (1 cycle), 6% (10 ² cycles), 4% (10 ⁵ cycles)
Thermal hysteresis	15 - 30°C
Bulk density	6,45 g/cm ³
Electrical resistivity	80 μΩcm (M) ; 100 μΩcm (A)
Corrosion resistance	Excellent
Melting point	1310°C
Elastic modulus	28 ÷ 41 GPa (Martensite), 83 GPa (Austenite)
Breaking load	1500 (cold worked) MPa



Thermomechanical testing

Shape Memory Effect set-up



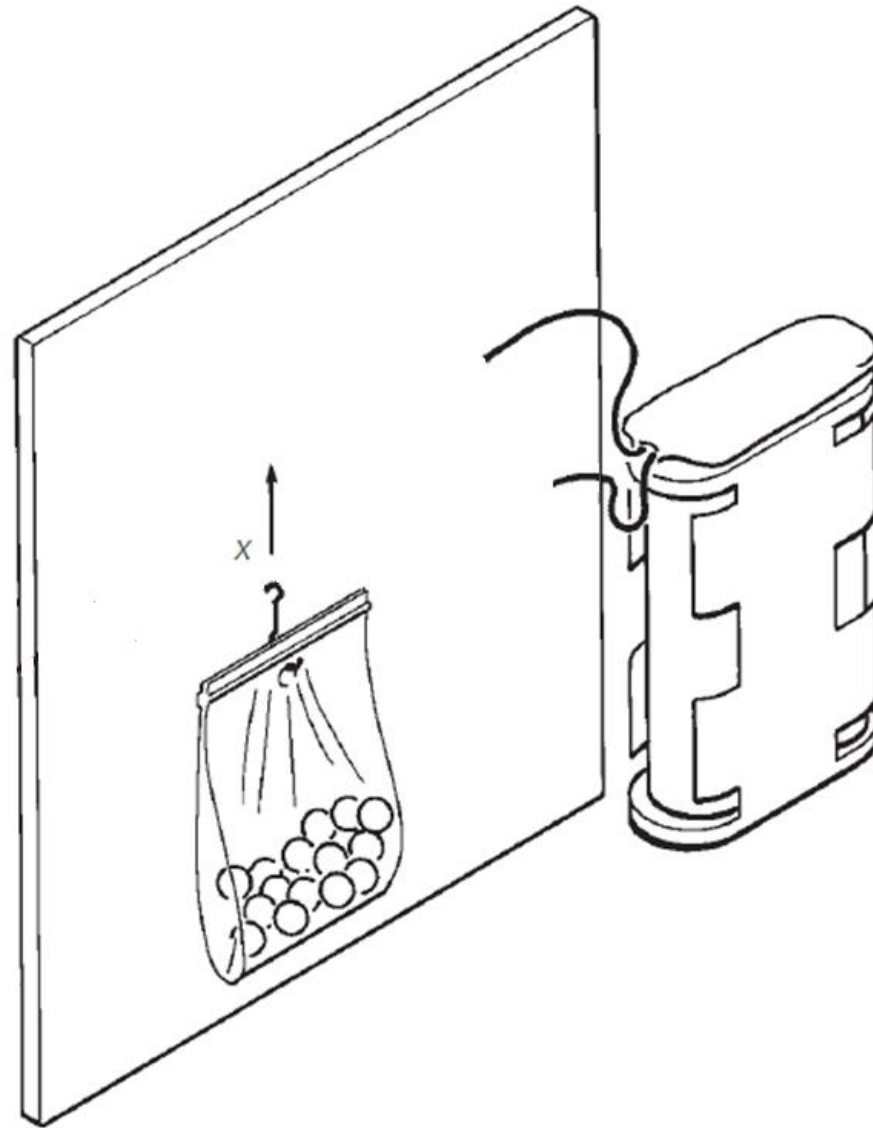
Actuator mechanical design principles

- Geometrical choice
- One-way or two-way memory effect
- Restoring force
- Anchoring points



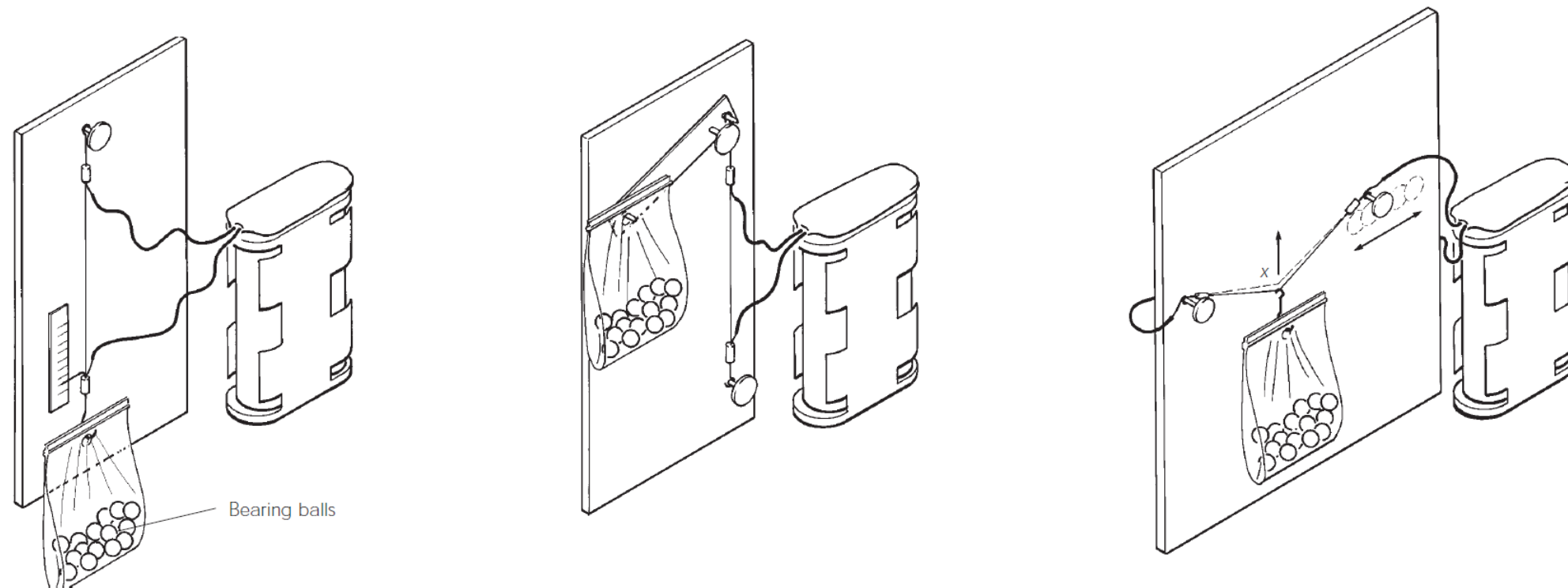
Actuator mechanical design principles

Geometrical choice



Actuator mechanical design principles

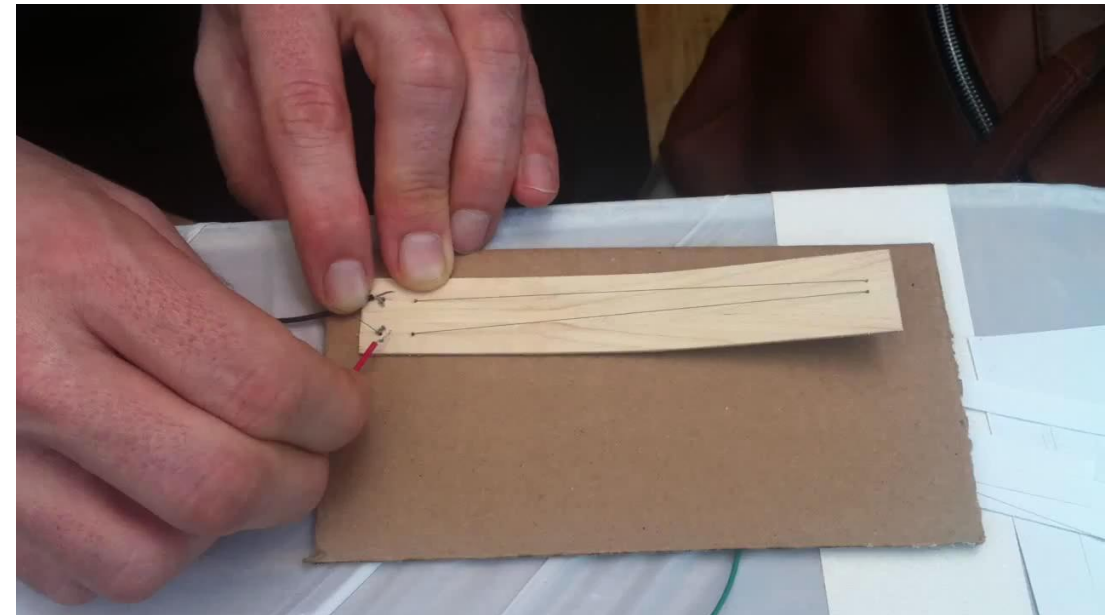
Geometrical choice



Actuator mechanical design principles

Geometrical choice

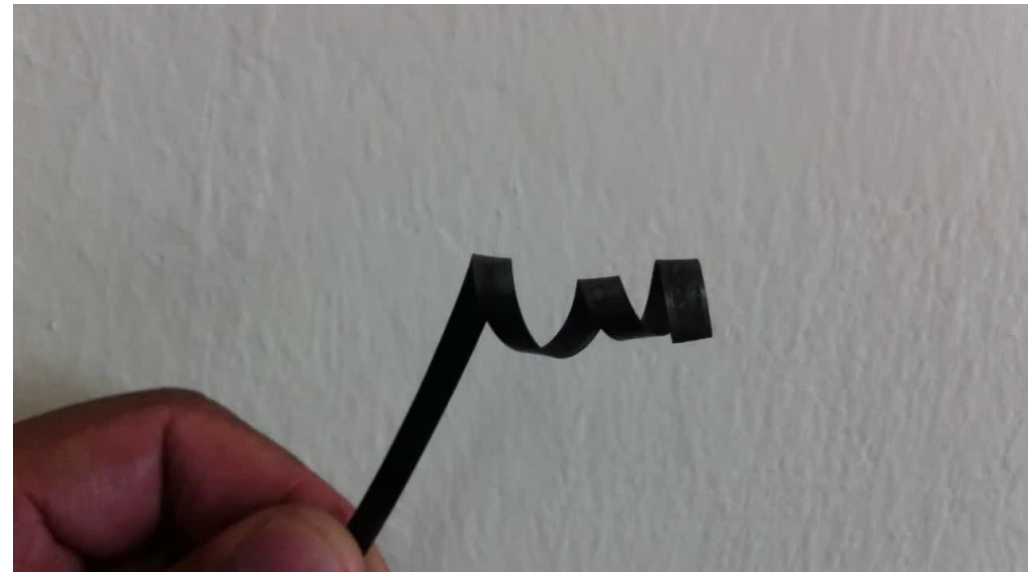
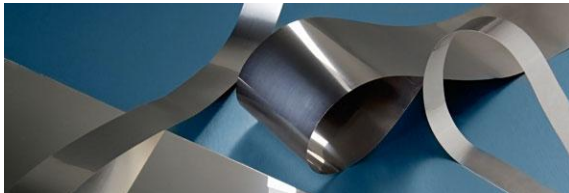
- Wires with different diameter and cross sections



Actuator mechanical design principles

Geometrical choice

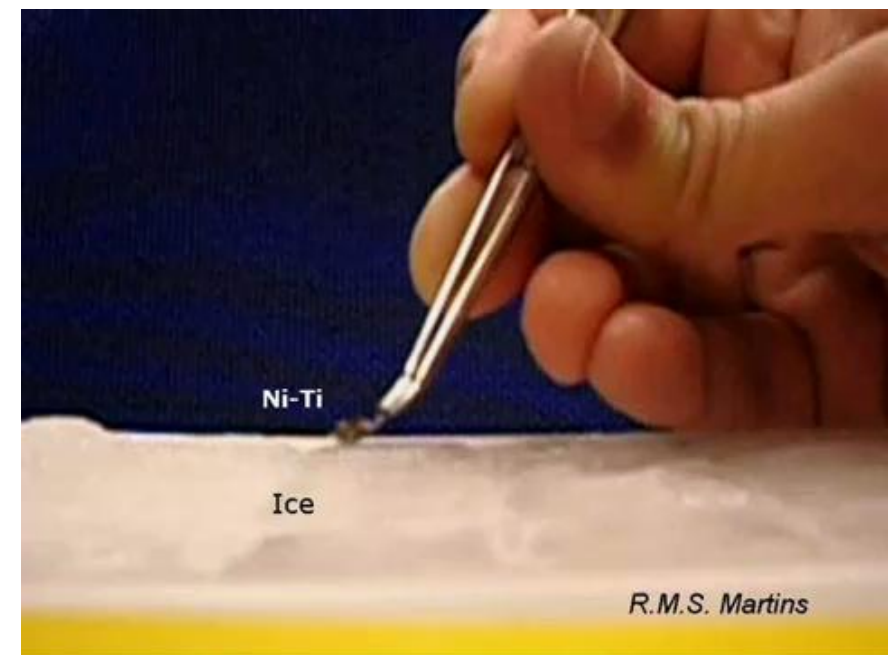
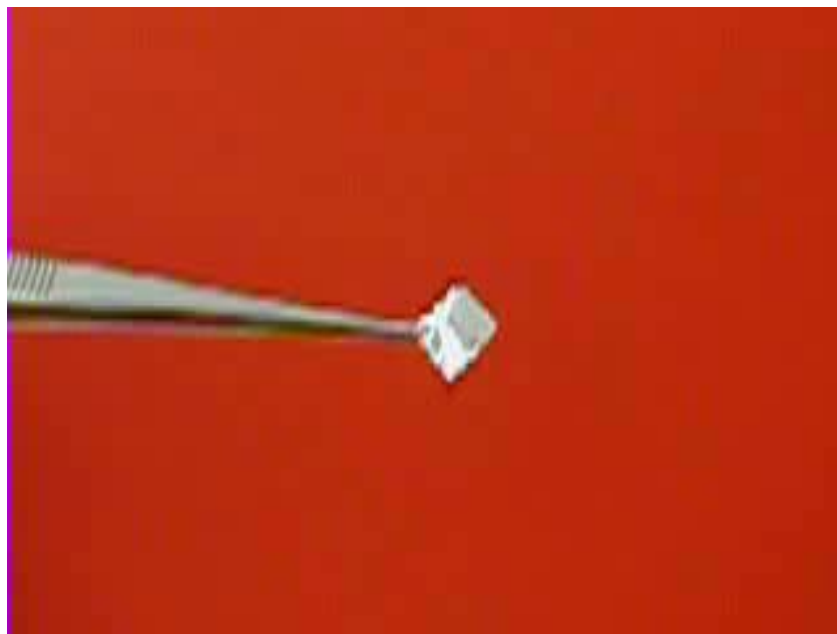
- Wires with different diameter and cross sections
- Ribbons with different geometrical proportions



Actuator mechanical design principles

Geometrical choice

- Wires with different diameter and cross sections
- Ribbons with different geometrical proportions
- Thin films / foils



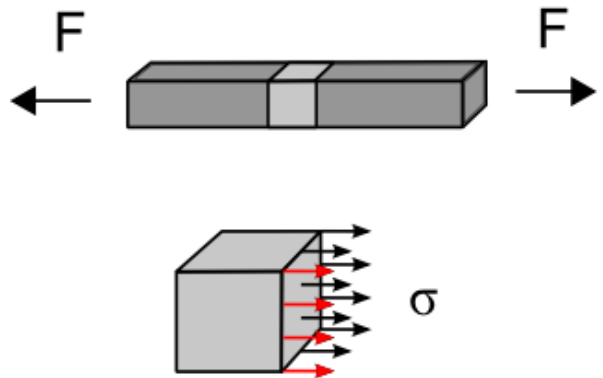
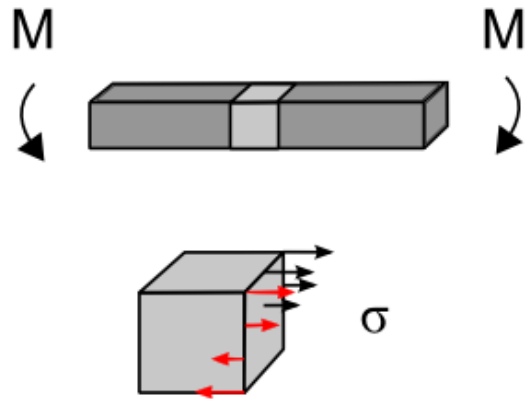
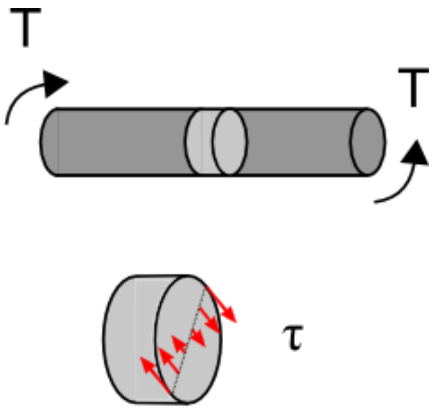
Actuator mechanical design principles

Geometrical choice



Actuator mechanical design principles

Mechanical load concepts

	Tensile deformation	Bending deformation	Torsion deformation
			
stress	$\sigma = \frac{F}{A}$	$\sigma_{(x)} = \frac{M_b x}{I}$	$\tau_{(r)} = \frac{M_t r}{J}$
efficiency	$\frac{\hat{\sigma}}{\sigma_{\max}} = 1$	$\frac{\hat{\sigma}}{\sigma_{\max}} = \frac{1}{2}$	$\frac{\hat{\tau}}{\tau_{\max}} = \frac{2}{3}$



Actuator mechanical design principles

Mechanical load concepts

Poisson's ratio, ν

the negative ratio of transverse to axial strain

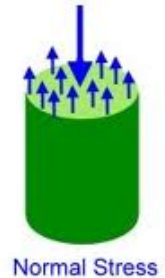
$$\nu = -\frac{d\varepsilon_{trans}}{d\varepsilon_{axial}} = -\frac{d\varepsilon_y}{d\varepsilon_x} = -\frac{d\varepsilon_z}{d\varepsilon_x}$$

the material's compressibility

Young Modulus, E

$$E = \frac{\sigma}{\varepsilon}$$

the material's response to linear strain



Shear Modulus, G

$$G = \frac{E}{2(1+\nu)}$$

the material's response to shear strain



Bulk Modulus, K

$$K = \frac{E}{3(1-2\nu)}$$

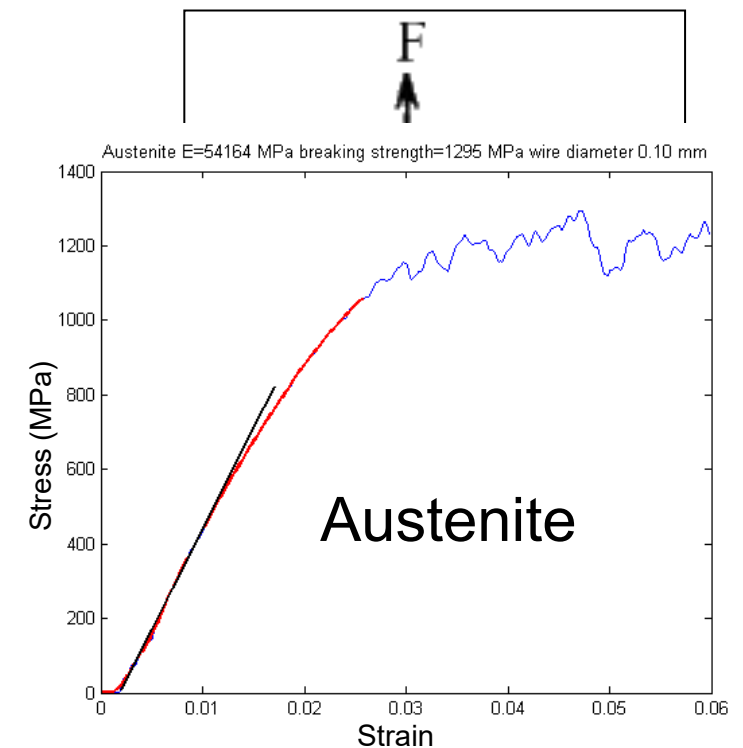
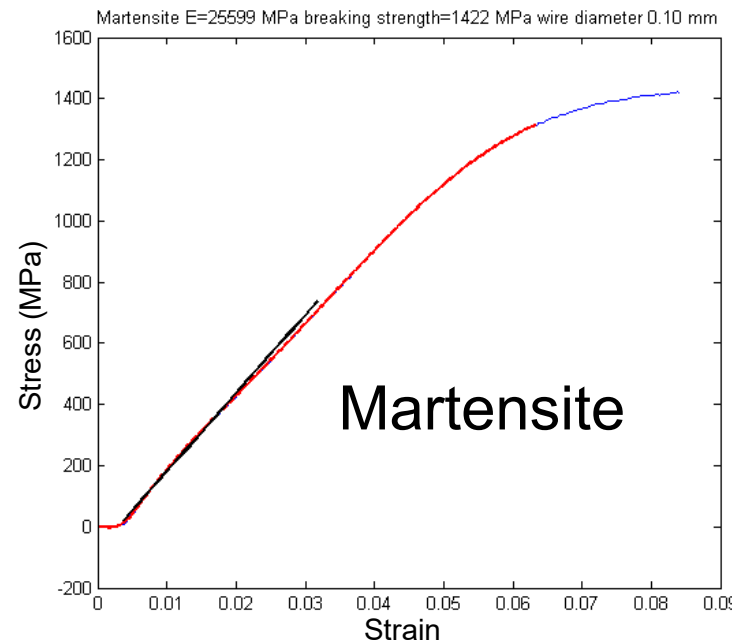
the material's response to uniform pressure



Actuator mechanical design principles

Geometrical choice – helicoidal springs

Wire diameter (d)
Spring diameter (D)
Number of coils (n)



Poisson's ratio (ν): 0.33

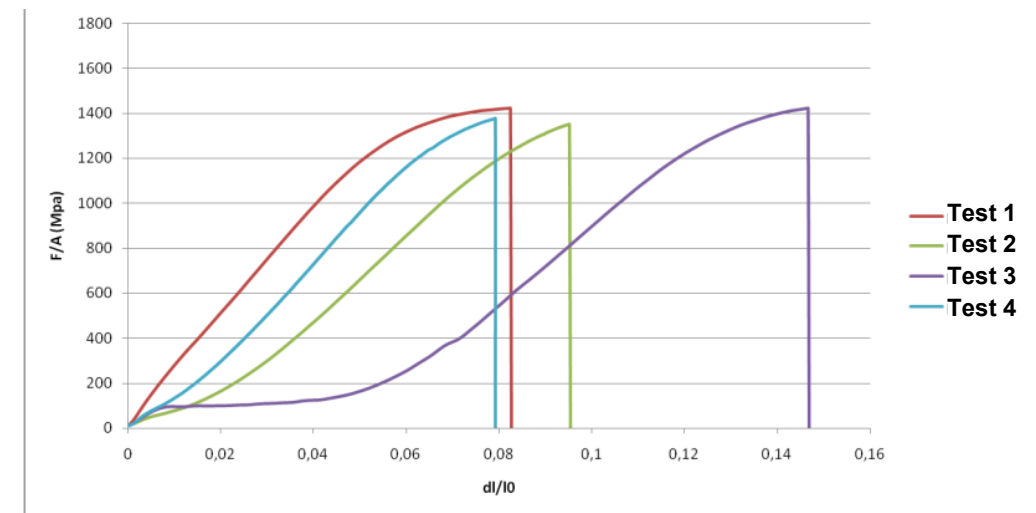
Austenitic Young Modulus (E_A): 55 GPa

Martensitic Young Modulus (E_M): 22 GPa

Ultimate stress (σ_r): 1400 MPa

Wahl's correction factor (c_f): 1.25

accounts for shear stress resulting from spring curvature



Breaking strength tests



Actuator mechanical design principles

Geometrical choice – helicoidal springs

Austenitic shear modulus:

$$G_A = \frac{E_A}{2(1+\nu)}$$

Martensitic shear modulus:

$$G_M = \frac{E_M}{2(1+\nu)}$$

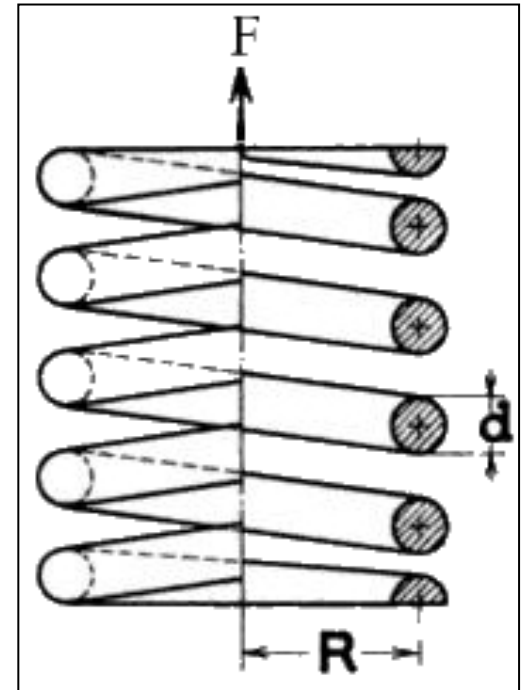
Spring index:

$$c = \frac{\hat{D}}{d}$$

Max shear stress:

$$\tau_{\max} = \frac{\sigma_r}{2c_f}$$

NOT LESS THEN 6!



Actuator mechanical design principles

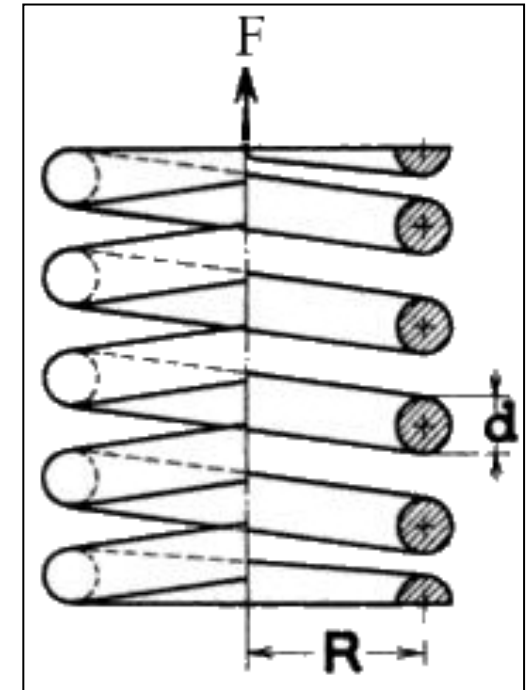
Geometrical choice – helicoidal springs

Austenitic spring rate:

$$K_A = \frac{G_A d}{8nc^3}$$

Martensitic spring rate:

$$K_M = \frac{G_M d}{8nc^3}$$



Spring free length: $L_0 = dn$

Max force: $F_{\max} = \frac{\tau_{\max} \pi d^3}{8D}$

Max HOT spring elongation:

$$\Delta x_{A_max} = \frac{F_{\max}}{K_A}$$

Max COLD spring elongation:

$$\Delta x_{M_max} = \frac{F_{\max}}{K_M}$$



Actuator mechanical design principles

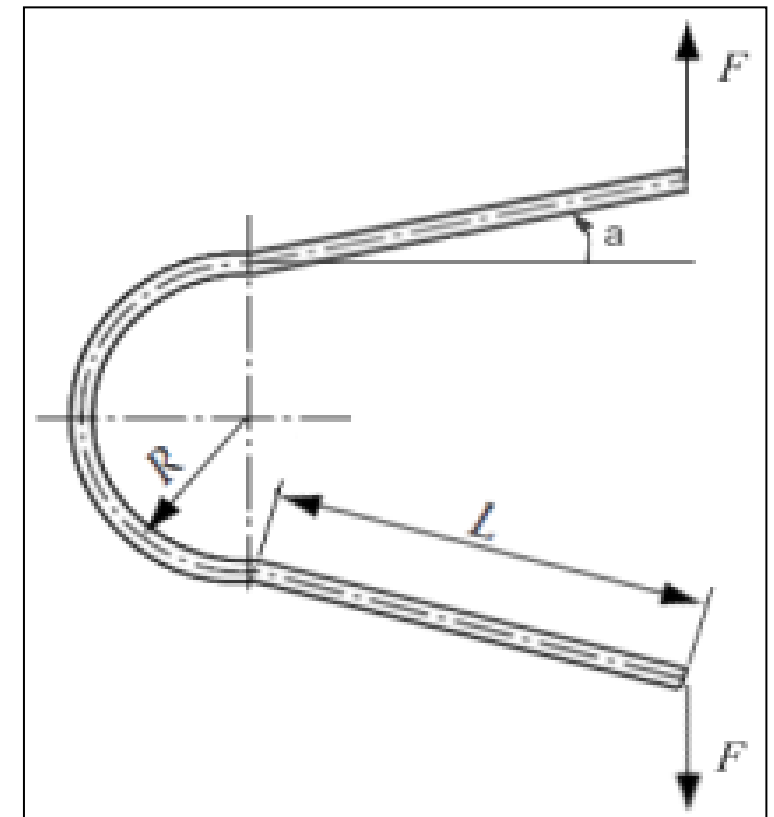
Geometrical choice – planar springs

Linear length (theoretical) - arm (L_t)

Wire diameter (d)

Number of U-elements (n)

Arm angle (α)



Poisson's ratio (ν): 0.33

Austenitic Young Modulus (E_A): 55 GPa

Martensitic Young Modulus (E_M): 22 GPa

Ultimate stress (σ_r): 1400 MPa

Wahl's correction factor (c_f): 1.25



Actuator mechanical design principles

Geometrical choice – planar springs

Real arm: $L = L_t \cos \alpha$

Half-circle radius: $R = d \frac{c}{2}$

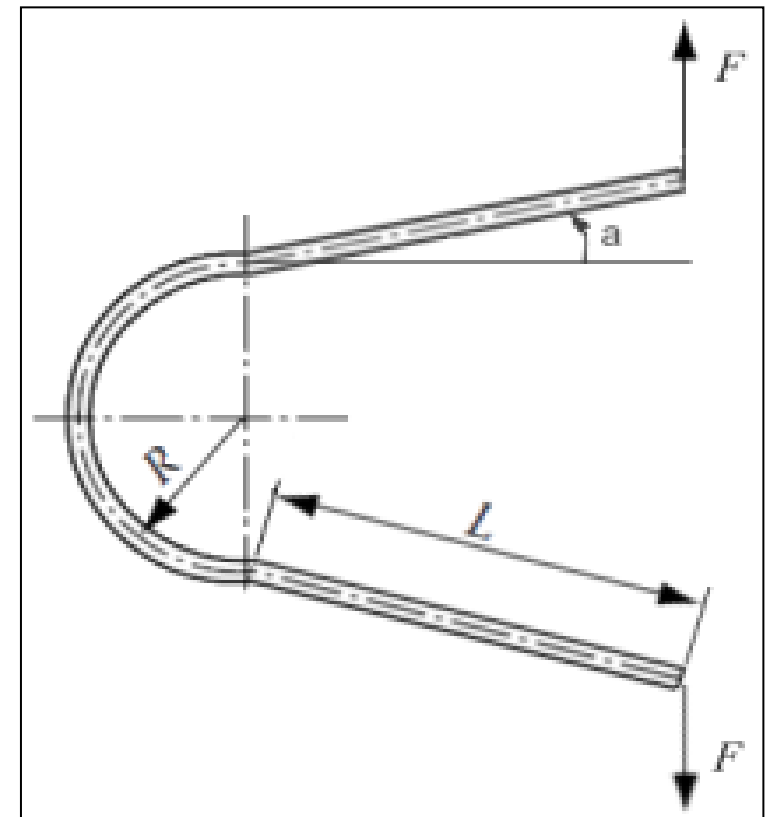
Spring width: $W = 2(L + R + D)$

Inertial moment: $I = \frac{\pi}{4} \left(\frac{d}{2} \right)^4$

Single element deflection: $y = \left(\frac{\pi F R^3}{2 E_A I} \right) + \left(\frac{\pi F R L^2}{E_A I} \right) + \left(\frac{2 F L^3}{3 E_A I} \right)$

Spring deflection: $y_{tot} = n y$

Max bending stress: $\sigma_{\max} = \frac{\sigma_r}{c_f}$

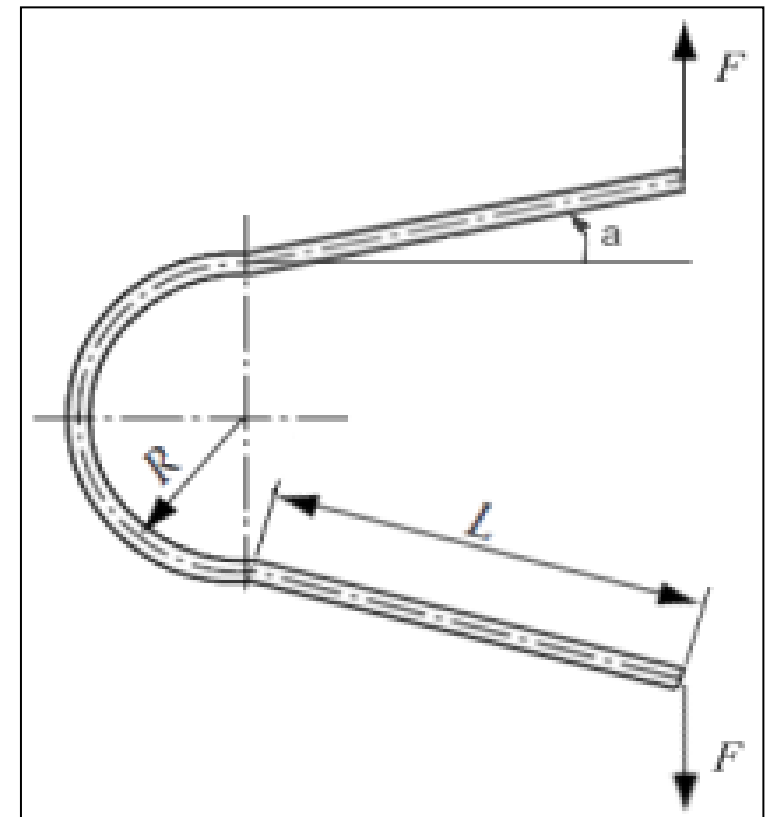


Actuator mechanical design principles

Geometrical choice – planar springs

Austenitic / Martensitic spring rate:

$$K_{A/M} = \frac{1}{n \left(\frac{\pi R^3}{2E_{A/M}I} + \frac{\pi RL^2}{E_{A/M}I} + \frac{2L^3}{3E_{A/M}I} \right)}$$



Spring free length: $L_0 = d(n+1) + 2nR + 2nL \sin \alpha$

Max HOT spring elongation: $\Delta x_{A_max} = \frac{F_{max}}{K_A}$

Max COLD spring elongation: $\Delta x_{M_max} = \frac{F_{max}}{K_M}$

Max force:

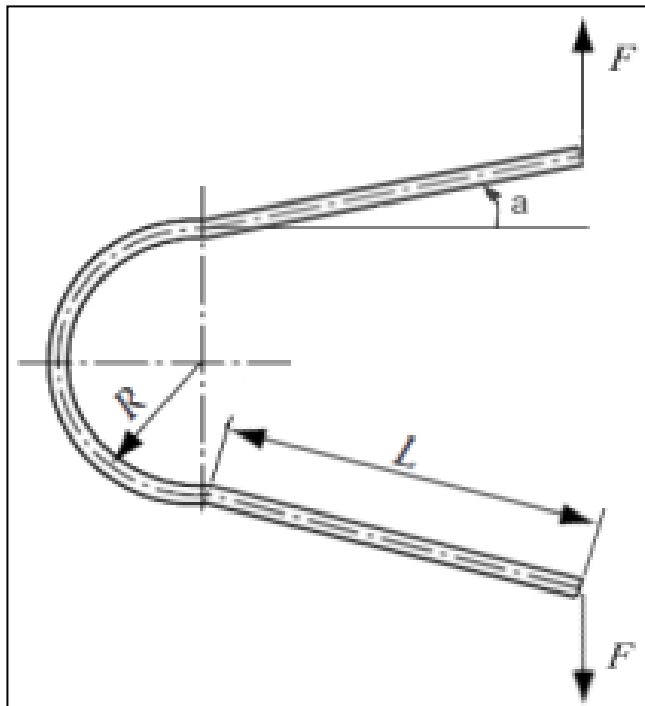
$$F_{max} = \frac{\sigma_{max} I}{(R + L_0) \frac{d}{2}}$$



Actuator mechanical design principles

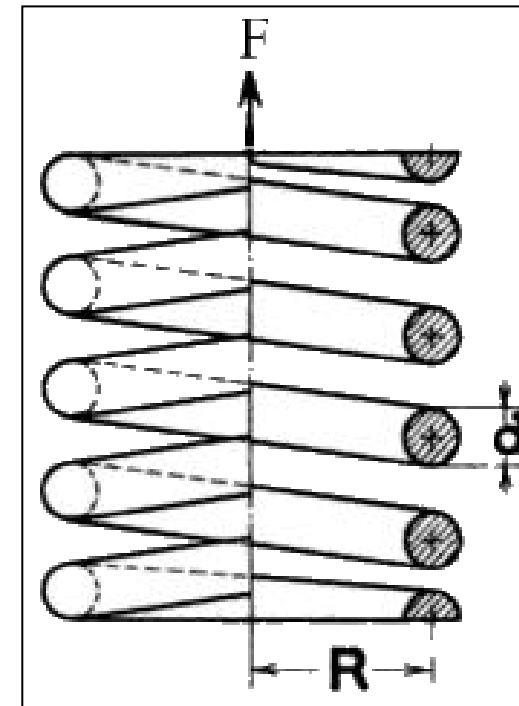
Geometrical choice

Planar springs



- more parameters can be adjusted
- practically 2D
- weaker

Helicoidal springs



- less parameters can be adjusted
- easier fabrication
- stronger



Actuator mechanical design

One-way or two-way memory effect

- Overdeformation while in the martensite phase
- Shape memory cycling (cool -> deform -> heat -> repeat)
- SuperElastic cycling (load -> unload -> repeat)
- Constrained temperature cycling of deformed martensite

Duering et al., 1990

NOTE: Recoverable strain decrease to 2%
Long-term fatigue data are missing
Stability performances not documented
Easy memory erasing (slight overheating: 250° C)

USE ONE-WAY SMA AND PROVIDE BIAS FORCE!



Actuator mechanical design

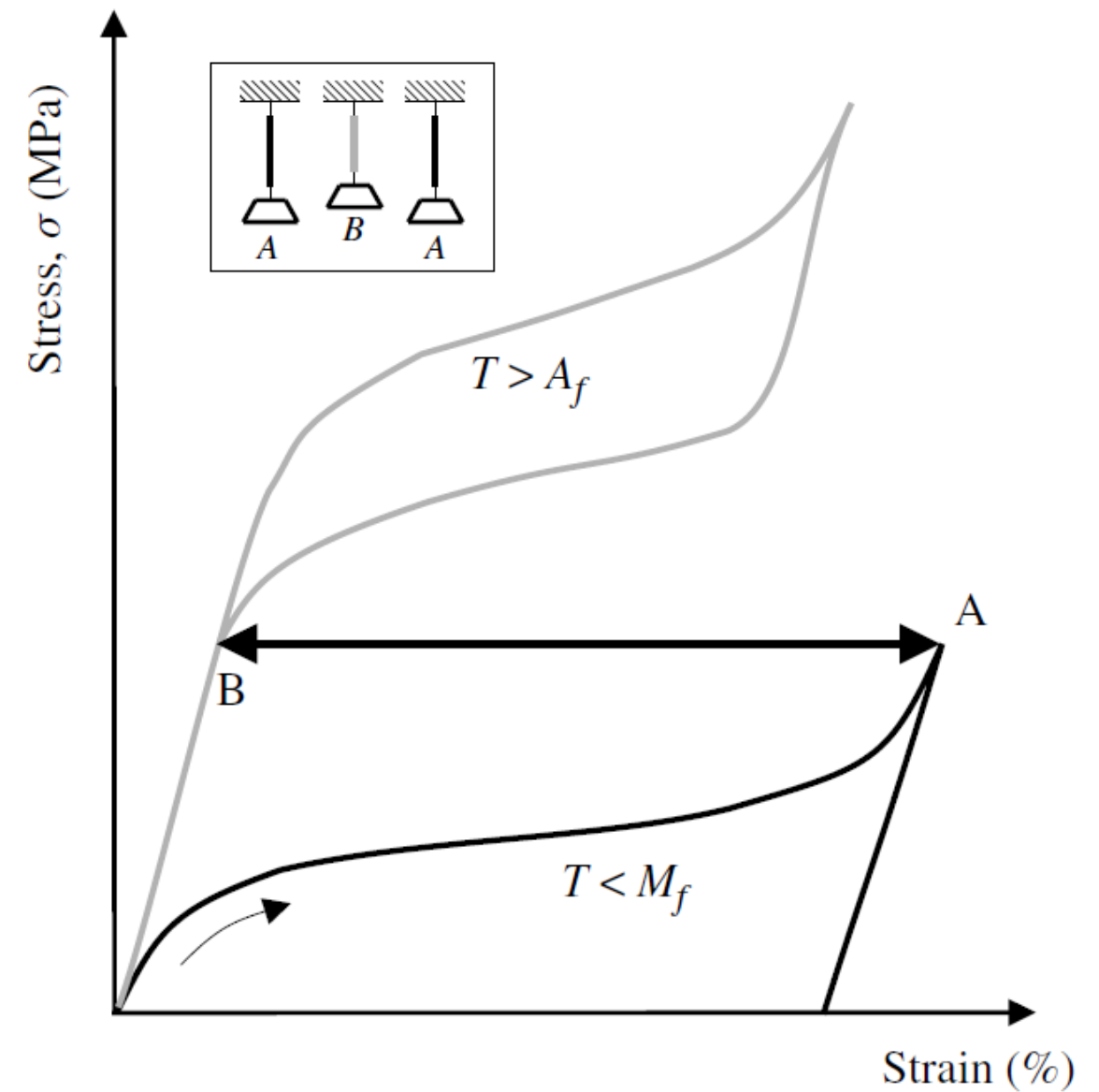
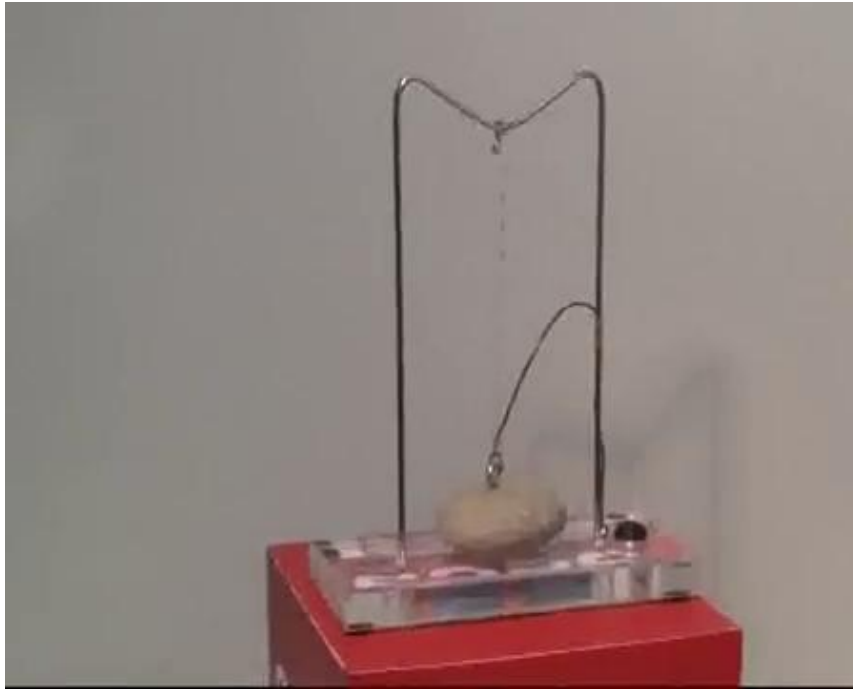
Restoring force

- Constant force
- Variable passive force
- Variable active force



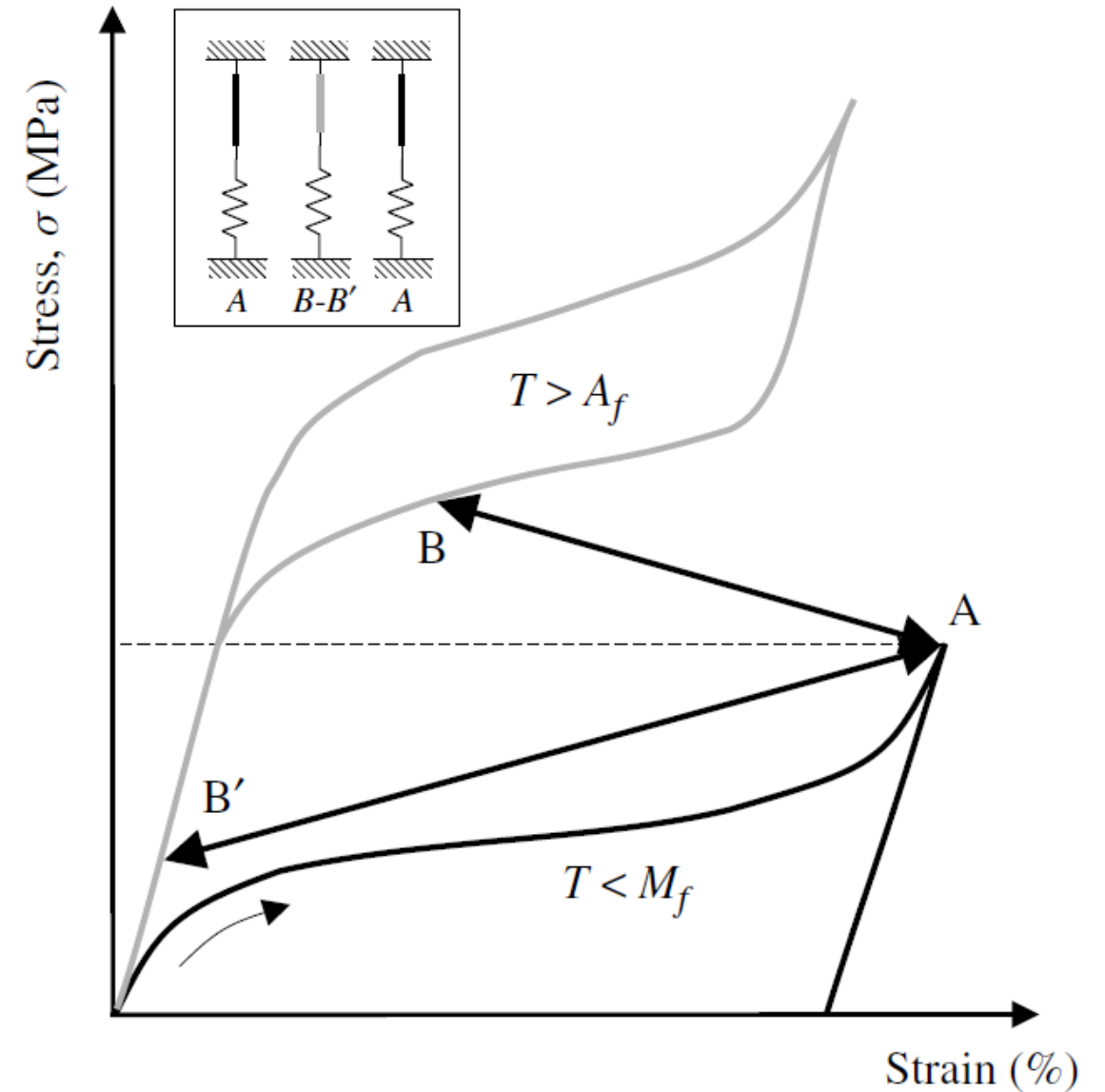
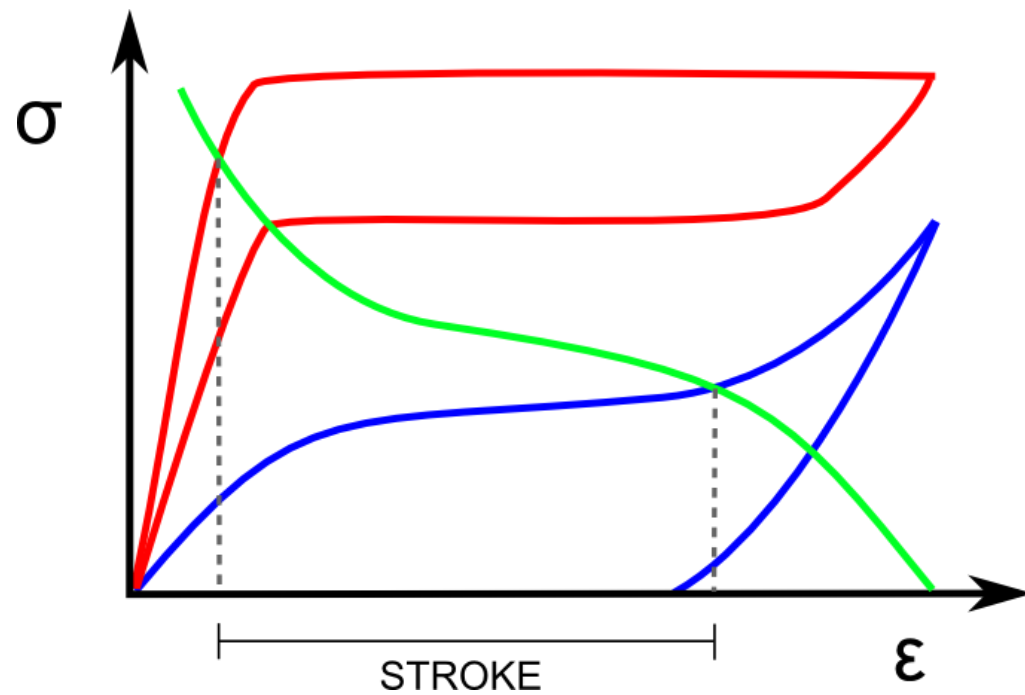
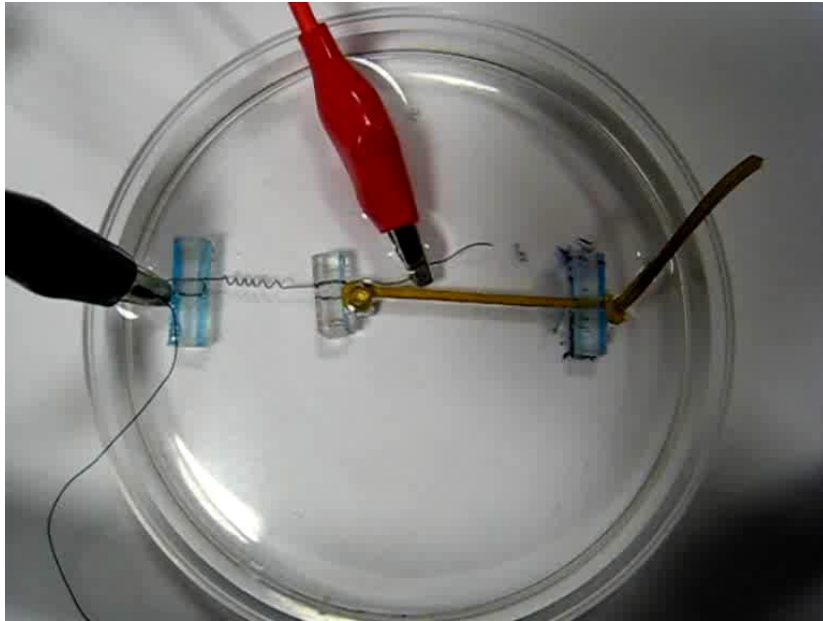
Actuator mechanical design

Restoring force - Constant force



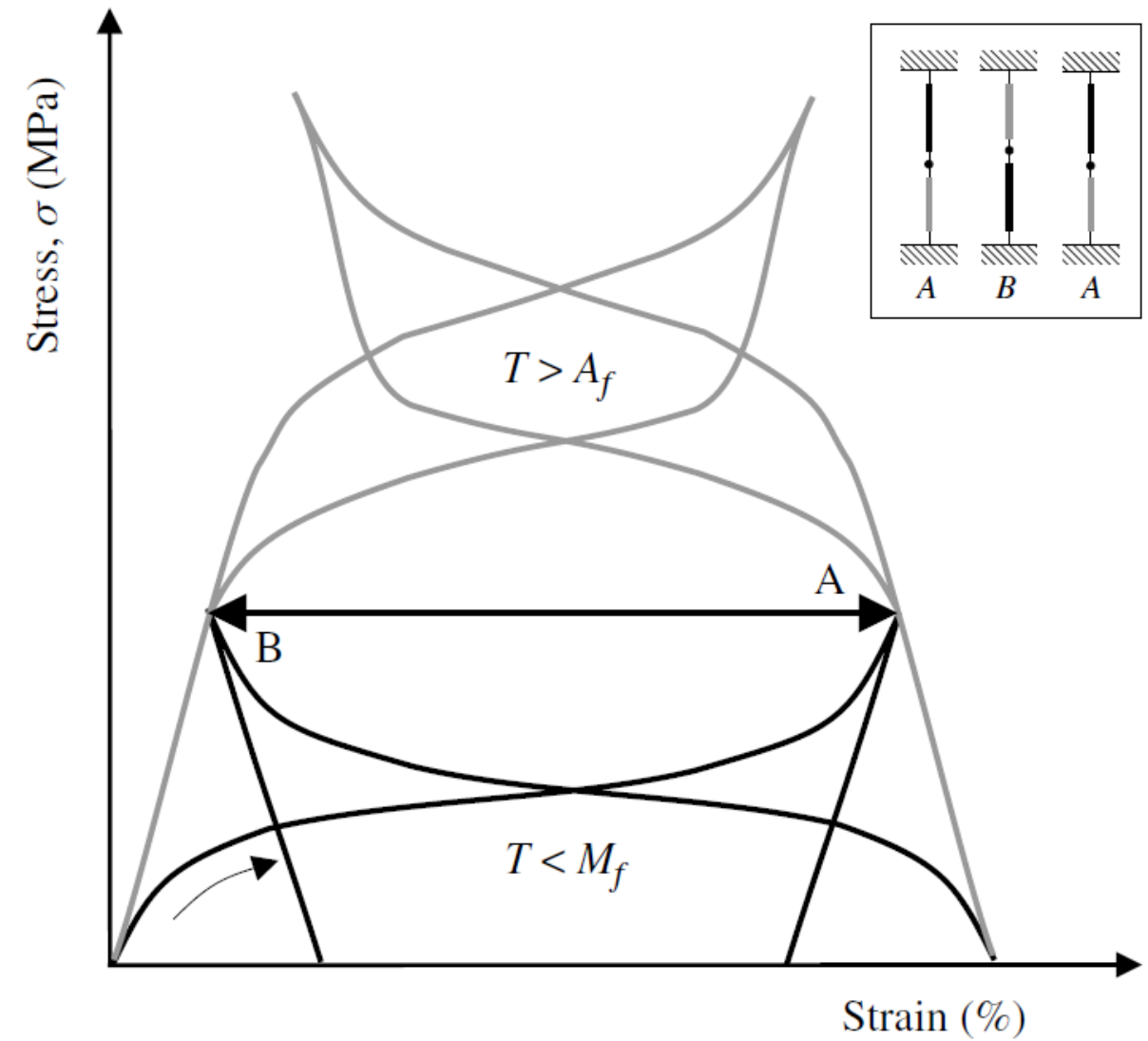
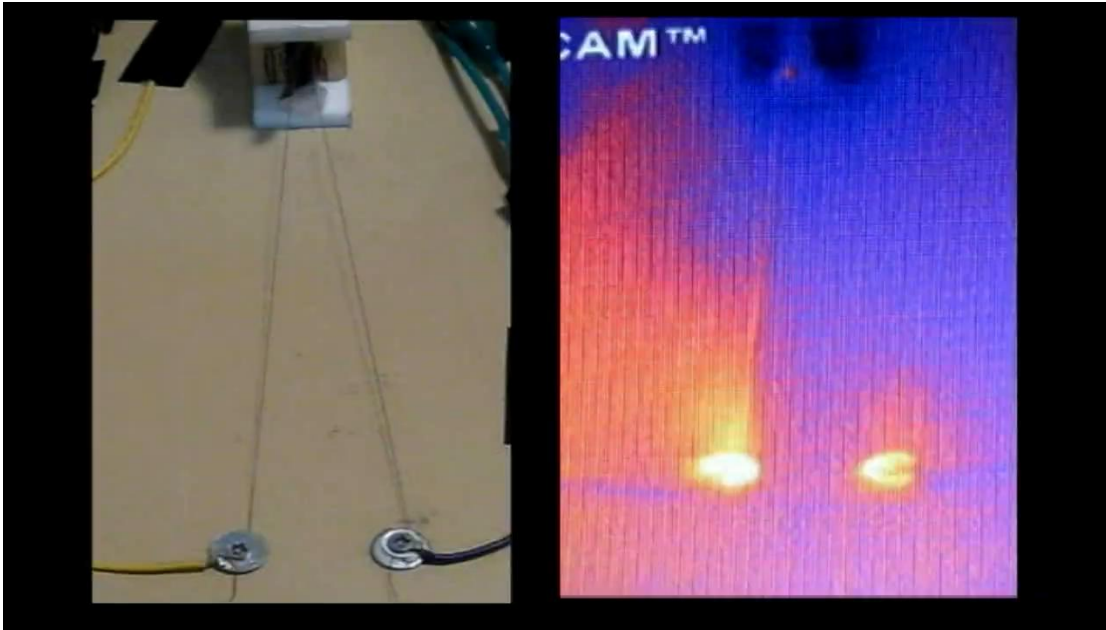
Actuator mechanical design

Restoring force - Variable passive force



Actuator mechanical design

Restoring force - Variable active force



Actuator mechanical design

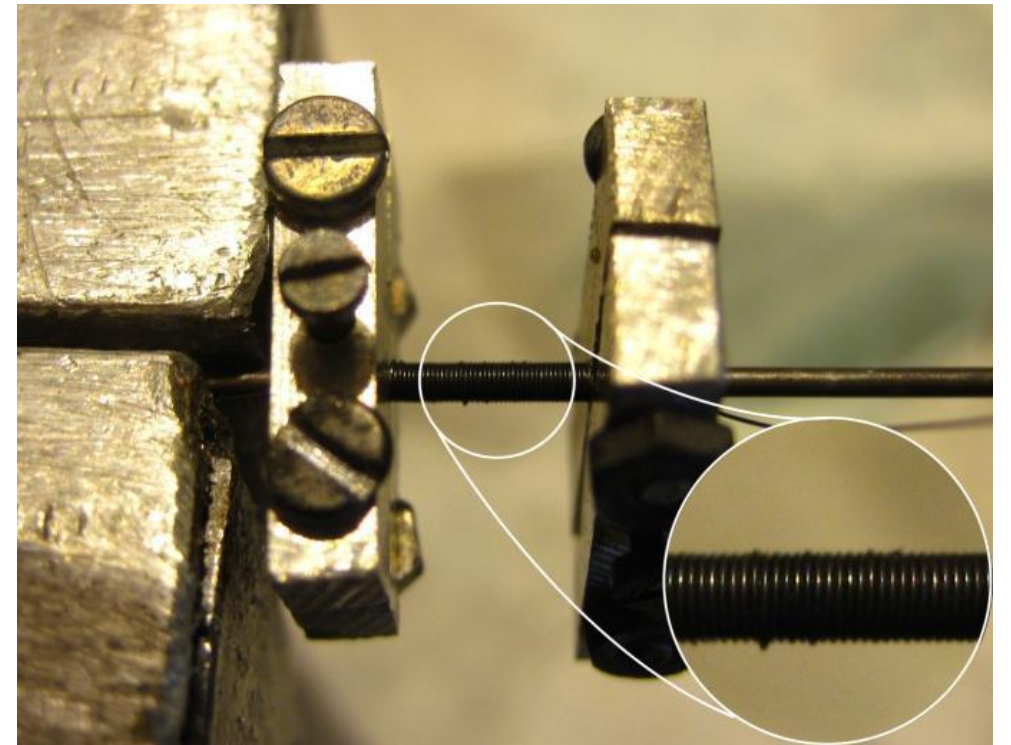
[SAES Getters SmartFlex Datasheet](#)



Practical use of SMAs

Forming (memorization)

1. Nitinol components are normally fabricated by holding the part in a fixture during heat treatment.
2. To memorize the desired shape, the spring has to be maintained at **450° C** for **30 minutes** under flowing **nitrogen** or other inert gases (to prevent oxidation).
3. After this time has elapsed, the spring has to be **water quenched**.



A good fabrication method is necessary to obtain repeatability among different actuator prototypes, since small differences and imprecision can deeply affect the mechanical performances of the actuators.



Practical use of SMAs

Forming (memorization) alternative

BUY THEM!

Nitinol Expansion Spring

[▶ Nitinol Index](#)



Nitinol Expansion Spring

When cool, spring can be extended to 14 cm (5-1/2"). Heated, contracts to 29 mm (1-1/8") overall. With 350 gram hanging mass shrinks from 60 mm (2-1/4") long coil (cool) to 30 mm (1-1/8") long (heated) with 2

Amps. 6 mm (1/4") coil, 750 μ m (.03") wire, activates at 45-55° C equivalent to (131-149° F). Use several in series for greater distance, or in parallel for greater force.

[PDF Manual](#)

Nitinol tension coil spring

Wire diameter 0.75mm (0.030in), mean coil diameter 5.55mm (0.22in), Alloy M

Analysis

- Wire diameter: 0.754 mm
- Mean coil diameter: 5.50 mm
- Total turns: 21.4
- Free length: 26.8 mm
- Austenite finish: 51.5°C
- Nickel: 55.36%
- Titanium: Balance

Item Niti-ES - \$44.95

 [add to cart](#)



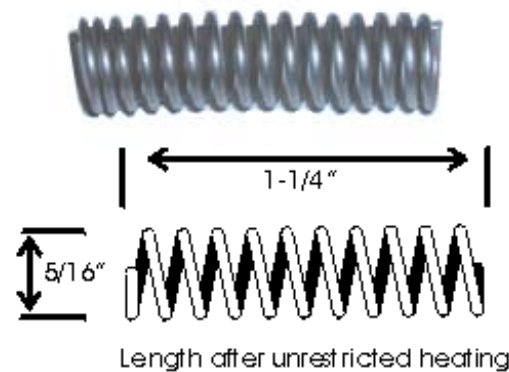
Practical use of SMAs

Forming (memorization) alternative

BUY THEM!

Nitinol Compression Spring

[▶ Nitinol Index](#)



Compression Spring When cool, can be pressed down to 16 mm (5/8"). Heated with 3 Amps, it extends to 30 mm (1-1/4") with over 4 Newtons of force! 8 mm (5/16") coil, 950 μ m (.04") wire, activates at 55-65° C equivalent to (131-149° F).

Nitinol compression coil spring

Wire diameter 1.0mm (0.039in), Mean coil diameter 7.0mm (0.27in), Alloy M

Analysis

- Wire diameter: 1.001 mm
- Mean coil diameter: 7.05 mm
- Total turns: 16.5
- Free length: 36.8 mm
- Austenite finish: 62.5°C
- Nickel: 55.14%
- Titanium: Balance

Item Niti-CS - \$44.95

 **add to cart**



Practical use of SMAs

Forming (memorization) alternative

BUY THEM!



QUOTAZIONE
Nr PQ111056.1-S

SAES Getters S.p.A.
Viale Italia, 77
20020 Lainate (Milano) Italy
Phone: +39 02 93178 1
Fax: +39 02 93178 320

A: Dr Matteo Cianchetti
Cliente: SCUOLA SUPERIORE SANT'ANNA
Cliente #: 0108163
Nazione: ITALIA
Dip:
Cc:
Saes Cc:

Da: Fabrizia Furlani
Data: 26-Oct-11
Tel: +390293178678
Fax: +390293178320
E-mail: Fabrizia_Furlani@saes-group.com

Egregio Dr Cianchetti,

Come da accordi con il nostro Ing. Butera, abbiamo il piacere di inviarLe la nostra migliore quotazione per le molle richieste:

	Descrizione Articolo	Codice	Quantità	Prezzo-EUR
1)	G/S/0.200/0.8d-200s-0.20p-C/S022	n.a	1 lotto da 10 pezzi (prezzo per lotto)	300.00/EA
2)	G/S/0.200/1.0d-200s-0.20p-C/S022	n.a	1 lotto da 10 pezzi (prezzo per lotto)	300.00/EA
3)	G/S/0.300/1.3d-150s-0.30p-C/S022	n.a	1 lotto da 10 pezzi (prezzo per lotto)	300.00/EA
4)	G/S/0.300/1.4d-200s-0.30p-C/S022	n.a	1 lotto da 10 pezzi (prezzo per lotto)	300.00/EA

EA:per unit/unitario TH:per thousand/al 1000



Control methods

Heavily dependent on the application

- on-off/modulated activation
- open/closed loop
- simple/complex geometry
- well known/unknown environment



Control methods

$$\rho c V \frac{dT}{dt} + \rho V \Delta H \frac{d\xi}{dt} = Ri^2(t) - hA(T - T_a)$$

ρ - mass density

c - specific heat

V - volume

H - latent transformation heat

ξ - martensitic fraction

i - electric current

R - electric resistance

h - Convection heat transfer coefficient

A - heat exchanging surface area

T_a - ambient temperature

Convective heat transfer correlations

Forced convection, Internal flow

Laminar flow $N_{uD} = 1.86(\text{RePr})^{1/3} \left(\frac{D}{L}\right)^{1/3} \left(\frac{\mu_b}{\mu_w}\right)^{0.14}$

Turbulent flow $h = \frac{k_w}{D_H} N_u$

- on-off/modulated activation
- open/closed loop
- simple/complex geometry
- well known/unknown environment

Natural convection, External flow

Vertical plate

$$h = \frac{k}{L} \left(0.68 + \frac{0.67 Ra_L^{1/4}}{(1 + (0.49 / \text{Pr})^{9/16})^{8/27}} \right)^2$$

Vertical cylinder

The same if $\frac{D}{L} \geq \frac{35}{Gr_L^{1/4}}$

Horizontal plate

$$h = \frac{k 0.54 Ra_L^{1/4}}{L} \quad 10^5 \leq Ra_L \leq 2 \times 10^7$$

$$h = \frac{k 0.14 Ra_L^{1/3}}{L} \quad 2 \times 10^7 \leq Ra_L \leq 3 \times 10^{10}$$

$$h = \frac{k 0.27 Ra_L^{1/3}}{L} \quad 3 \times 10^5 \leq Ra_L \leq 10^{10}$$

Horizontal cylinder

$$h = \frac{k}{D} \left(0.6 + \frac{0.38 Ra_D^{1/6}}{(1 + (0.559 / \text{Pr})^{9/16})^{8/27}} \right)^2$$



Control methods

$$\rho c V \frac{dT}{dt} + \rho V \Delta H \frac{d\xi}{dt} = Ri^2(t) - hA(T - T_a)$$

ρ - mass density

c - specific heat

V - volume

H - latent transformation heat

ξ - martensitic fraction

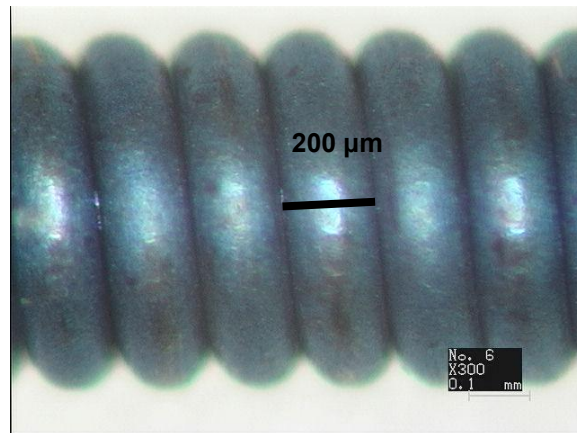
i - electric current

R - electric resistance

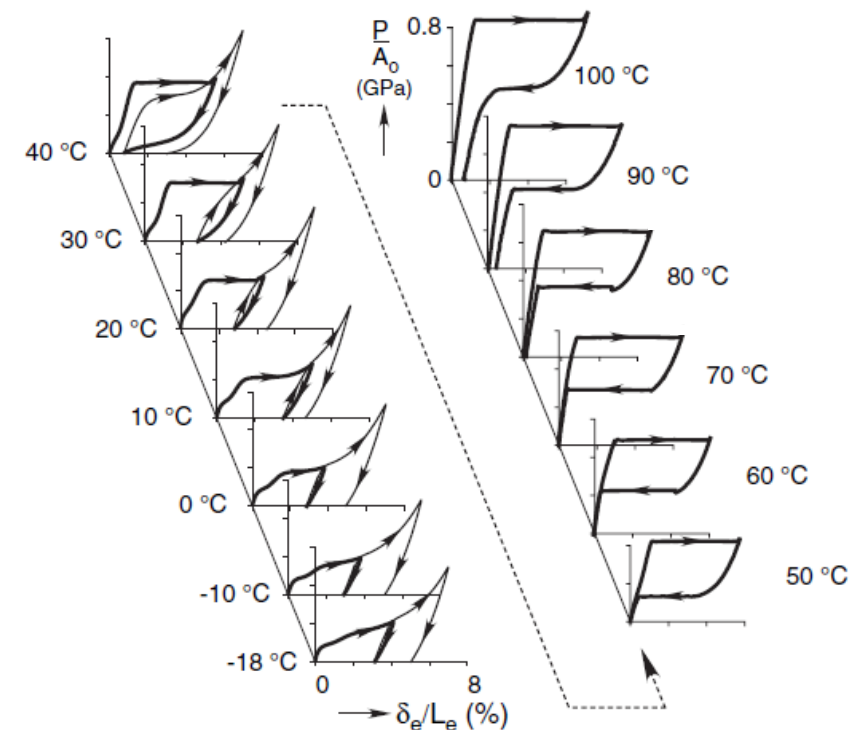
h - Convection heat transfer coefficient

A - heat exchanging surface area

T_a - ambient temperature



- on-off/modulated activation
- open/closed loop
- simple/complex geometry
- well known/unknown environment



Finite Element Analysis



1. The springs could be passively deformed by external forces or antagonistic actuators so that the stress exceeds the maximum allowed value.
2. We can use the stress induced Martensite to auto-limit the stress the spring experiences.



Control methods

The proportional control of SMAs based on mathematical models have often not sufficient accuracy

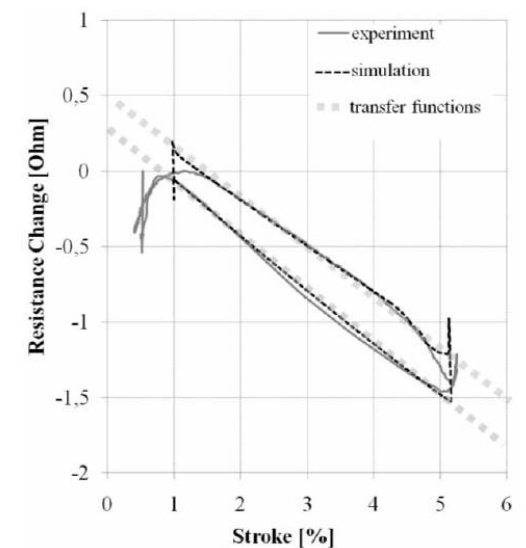
- on-off/modulated activation
- open/closed loop
- simple/complex geometry
- well known/unknown environment

Two available methods:

- External variables feedback
 - displacement
 - force
 - Internal state variable feedback
 - temperature
 - electric resistance
- likely to cause overheating
- difficult to apply in micro-scale application

The detection of inner electrical resistance allows to detect the alloy martensite fraction. The method consists in measuring the electrical resistance of an SMA element, calculating a maximum safe heating current as a function of measured resistance, and ensuring that the actual heating current does not exceed this maximum value.

Limit: SMA resistance is also sensitive to the deformation!!



Control methods

Electric resistance feedback

$$\frac{dR}{R} = \pi_e d\sigma + K_\varepsilon d\varepsilon + \alpha_{RT} dT$$

Where: T = temperature; σ = stress; ε = strain; π_e = piezoelectric coefficient

R = electric resistance; α_{RT} = coefficient; K_ε = shape sensitivity coefficient

- on-off/modulated activation
- open/closed loop
- simple/complex geometry
- well known/unknown environment

You don't know if the resistance change is due to a stretch (external stress) or to a temperature change (environment).

- Known stress: through calibration curves R(T) the resistance is sufficient to derive the deformation
- Unknown stress: stress measurement is needed together with resistance

Advantages:

- Avoiding overheat of SMA
- Robustness against changes of heating and cooling



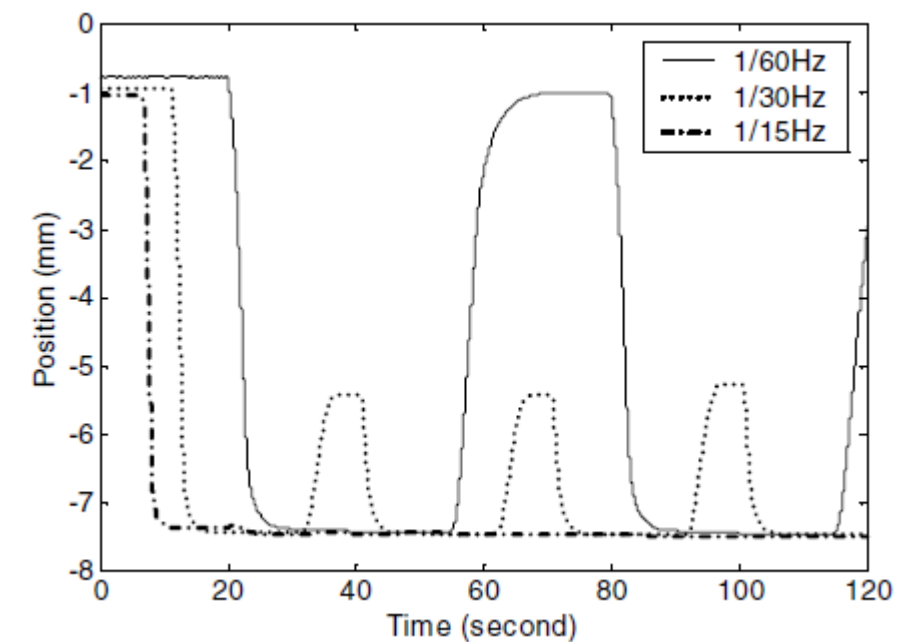
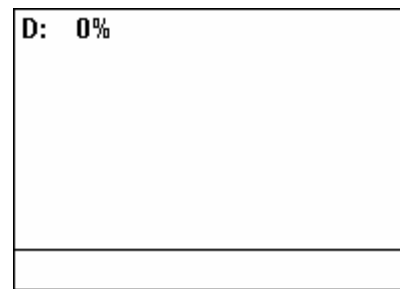
Driving methods

- Continuous PID controller
- PWM PID controller
 - Duty cycle
 - Frequency

$$RMS = A_{\max} \sqrt{D}$$

$$D = \frac{\tau}{T}$$

- τ - Duration of activation
- T - Period

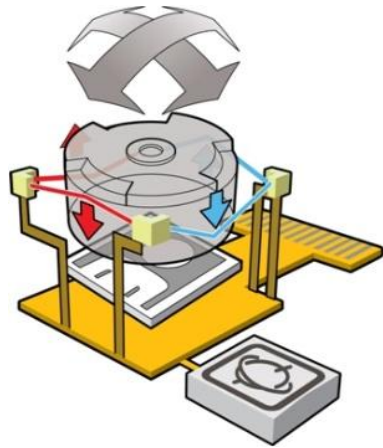


PWM has the advantages of being **robust to disturbances**, effective in **saving energy** and **easily implemented** using microprocessors

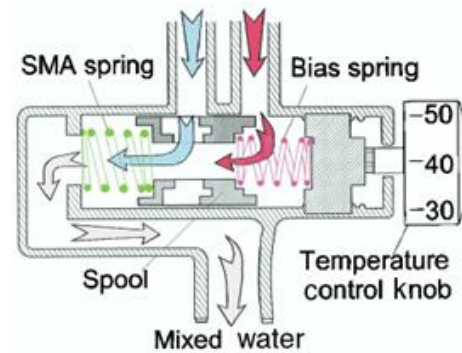


Applications

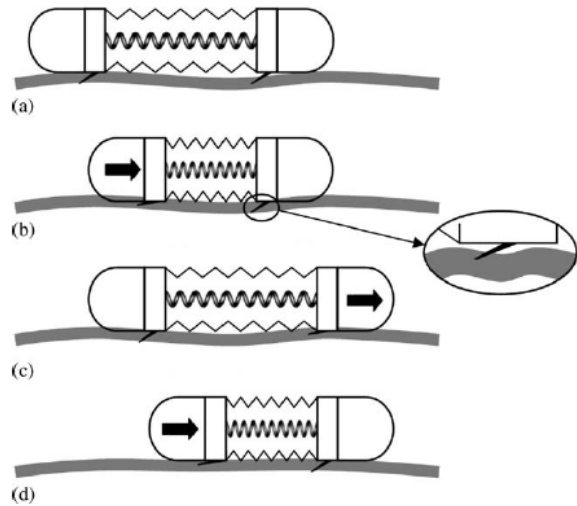
Examples



Optical image stabilization in a smartphone miniature camera



Thermostatic mixing valve

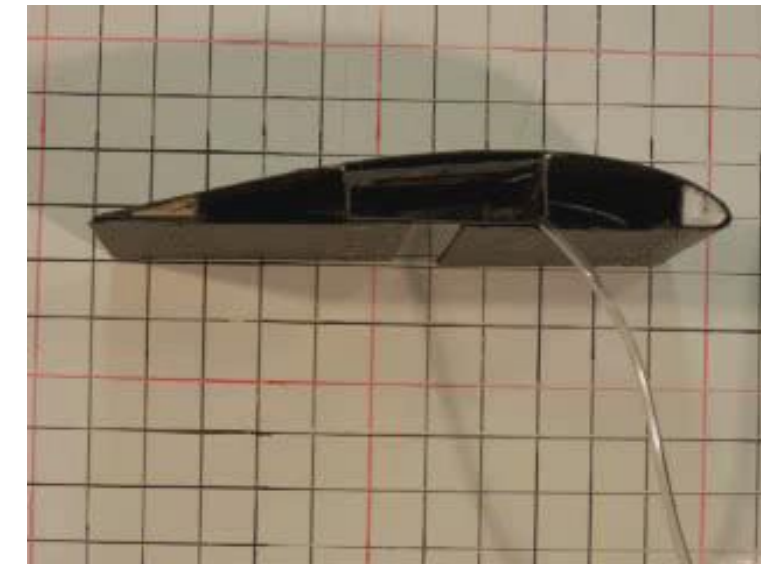
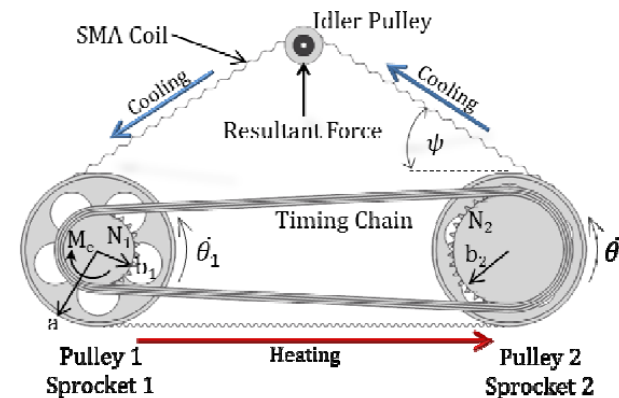


Earthworm-like micro robot

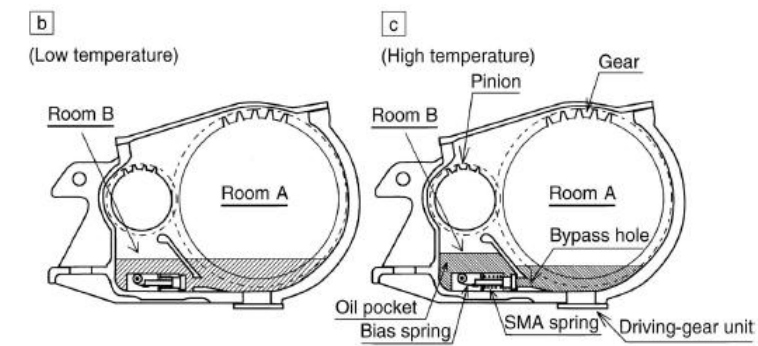


Loud speaker

Heat engine



Morphing Wing with high force actuator



Automatic oil-level-adjustment device for the Shinkansen bullet train driving-gear unit



Soft robotics applications

Peristaltic locomotion

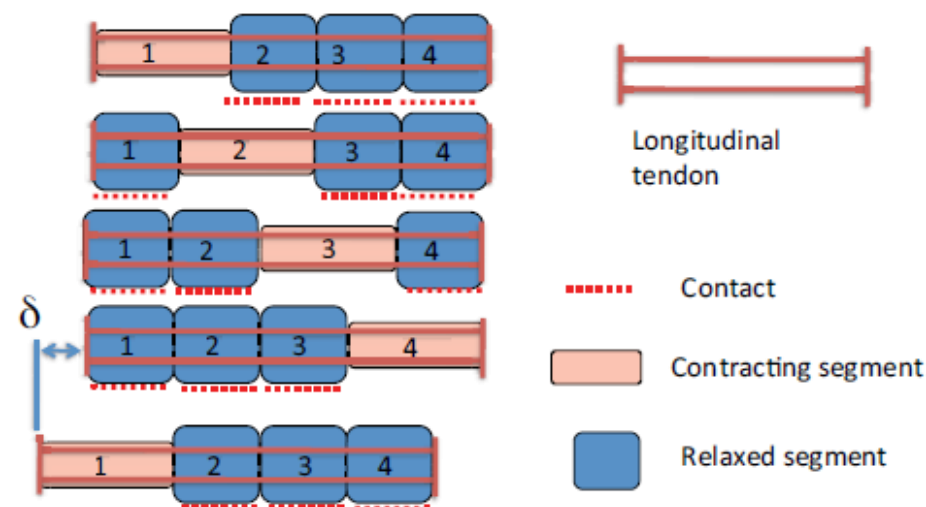
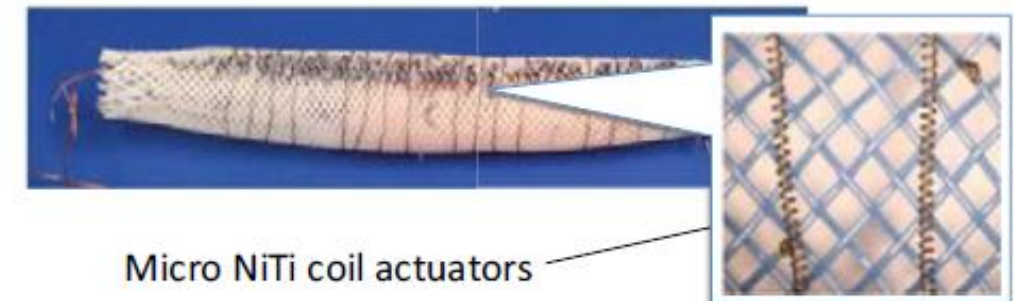
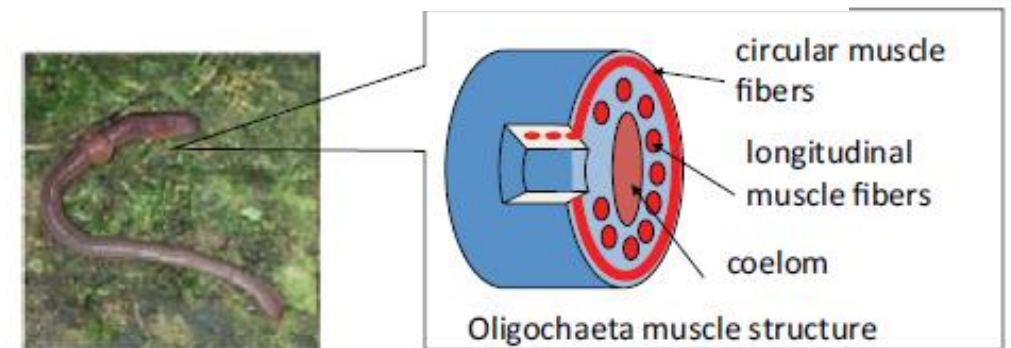
Bioinspired robotic earthworm composed of:

- flexible braided mesh-tube structure as a support
- SMA springs arranged circularly to squeeze

Sequential activation of the spring groups leads to **peristaltic motion** that can be used as a locomotion system.

400 mA current with **short pulse** time to activate the springs and the coordination of the segments is obtained by a **purposely developed gait controller**.

Closed loop control is implemented with the measure of the radius changes by means of **Hall effect sensors**.



Peristaltic locomotion with
antagonistic actuators in soft robotics

S.-O. Seok, C. Onal, R.J. Wood, D. Rus, and S. Kim

SEOK S., ONAL C.D., WOOD R., RUS D., KIM S., "Peristaltic Locomotion with Antagonistic Actuators in Soft Robotics" ICRA'10, Anchorage, May 3-8 2010, Alaska, p. 1228-1233.



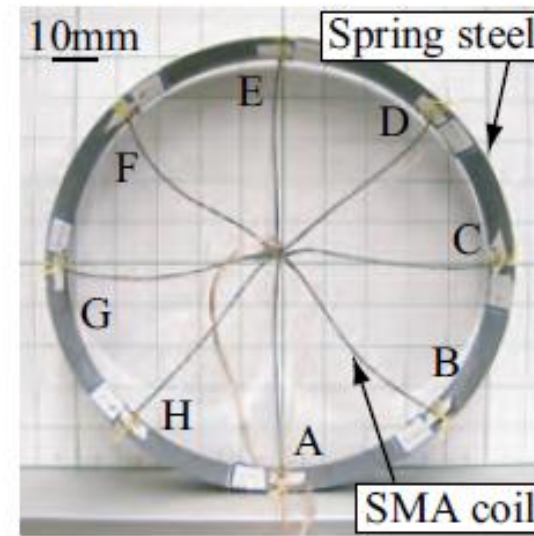
Soft robotics applications

Rolling robot

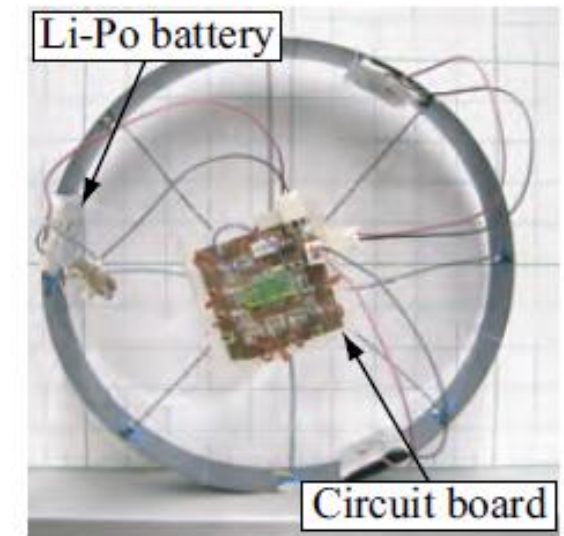
The robot is basically composed of:

- a deformable circular shell
- eight SMA springs radially arranged

When the springs contract the circle transforms into an **ellipse** becoming instable so that gravitational potential energy generates a moment around the point of contact between the body and the ground. This causes a **body roll** until it achieves a new stable state.

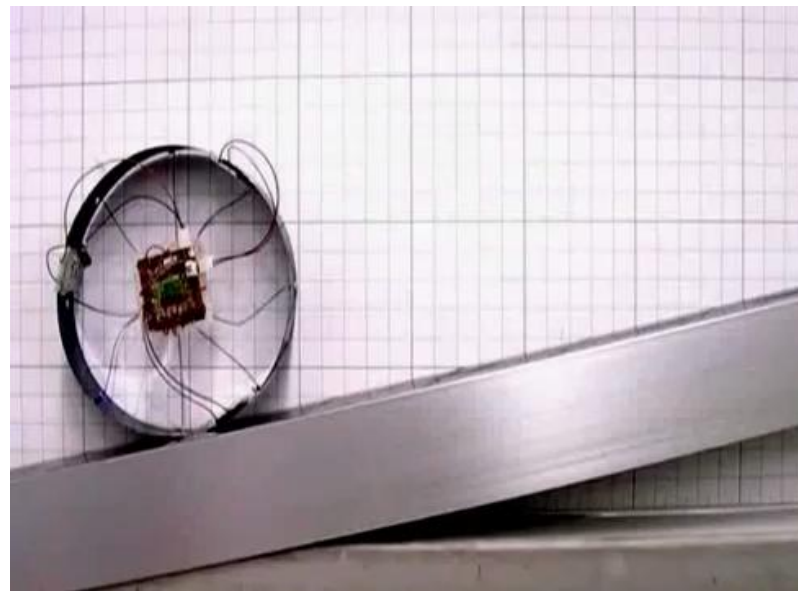
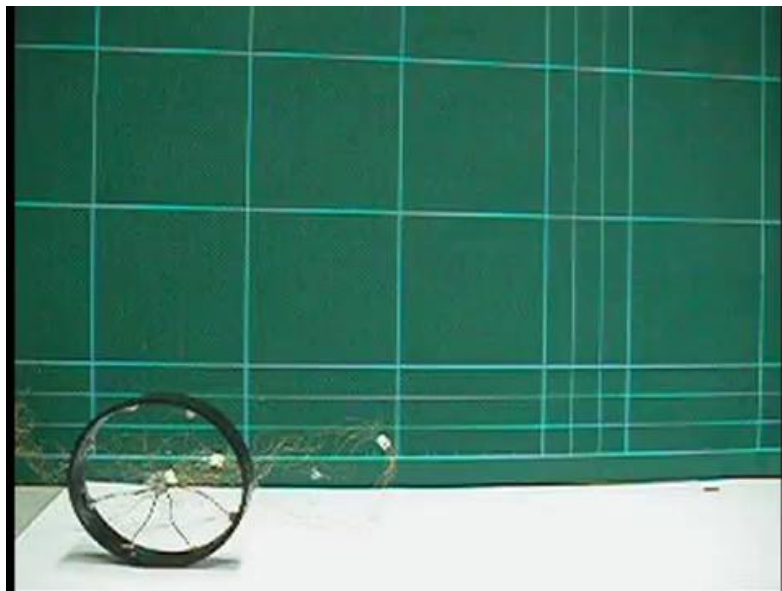


(a) Robot I



(b) Robot II

MATSUMOTO Y., NAKANISHI H., HIRAI S. "Rolling locomotion of a deformable soft robot with built-in power source" CLAWAR'08, Coimbra, September 8- 10, 2008, Portugal, p. 365–372.



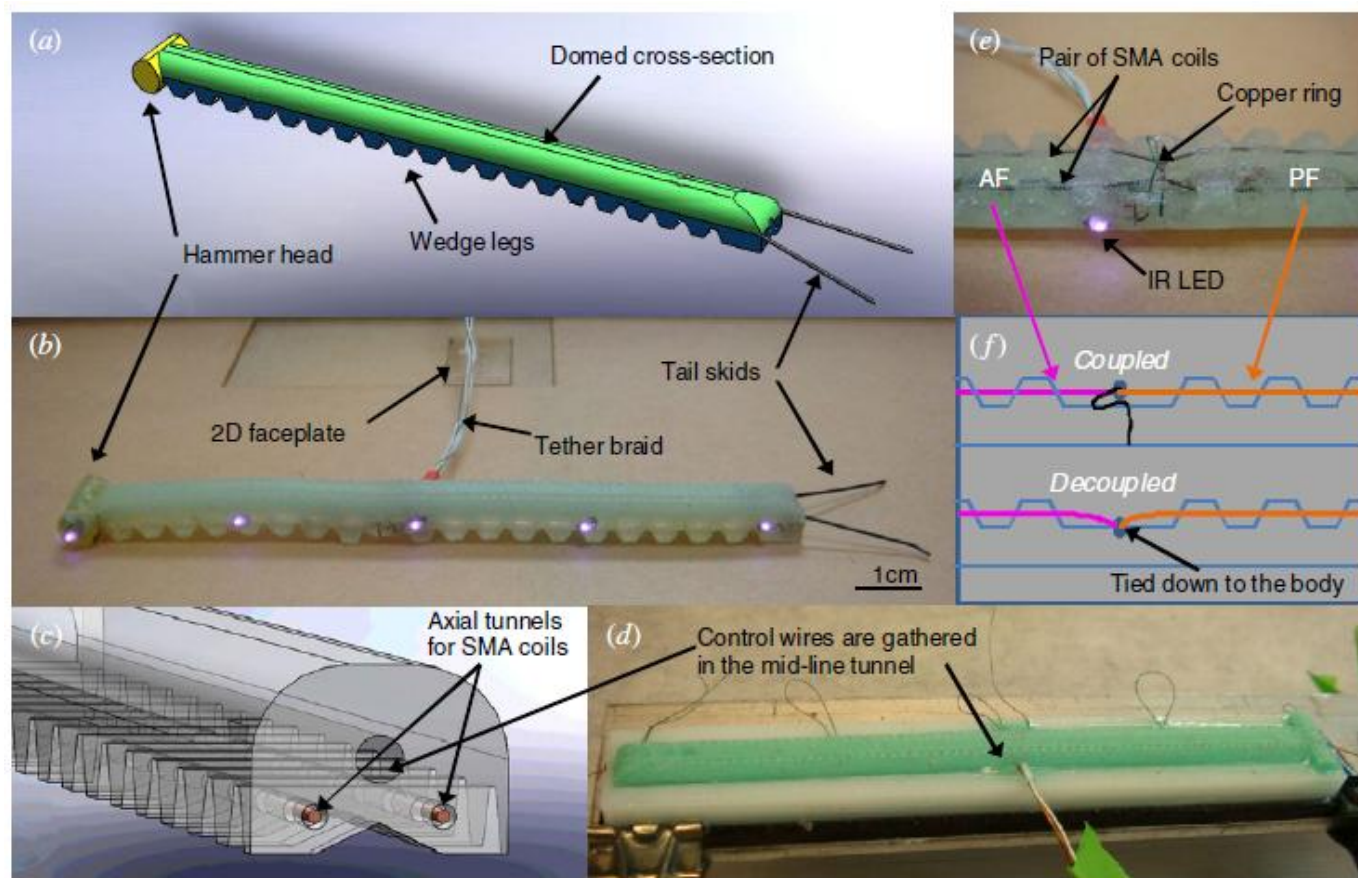
The provided activation is **on/off** through a **PWM** and with the appropriate voltage pattern a **continuous roll** is achievable



Soft robotics applications

GoQBot

It replicates a very fast self-propelled wheeling behavior producing a **ballistic rolling** by changing its body configuration in 250 ms.



LIN H.T., LEISK G.G., TRIMMER B., "GoQBot: a caterpillar-inspired soft-bodied rolling robot" *Bioinspiration and Biomimetics*, vol. 6, 2011, 026007.



Functional components:

- a soft silicone body molded in a straight and long geometry (bar-like)
- two independent SMA actuators fixed in parallel respect to the body allowing its anterior and posterior flexion

The control system is based on **PWM** driving with **high power stimulations** for very impulsive response

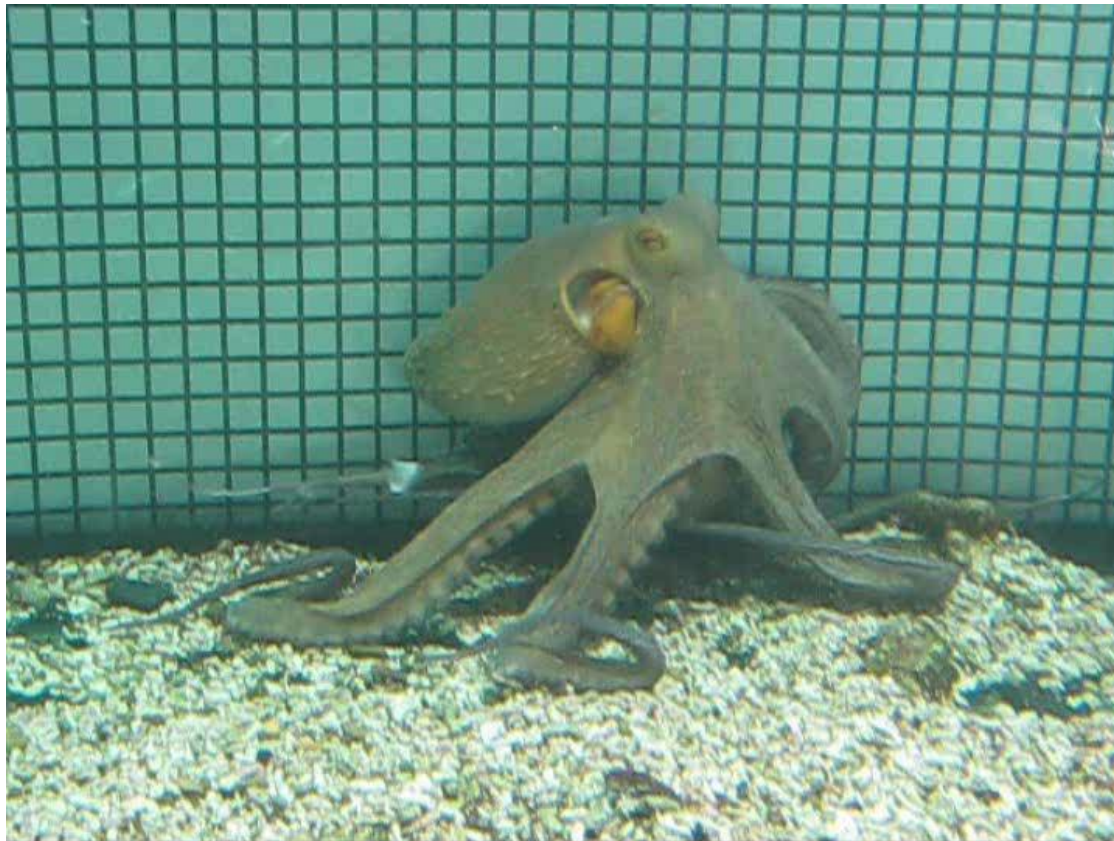


Case study



OCTOPUS IP (2009-2013)

Novel Design Principles and Technologies
for a New Generation of High Dexterity
Soft-bodied Robots Inspired by the
Morphology and Behaviour of the
Octopus

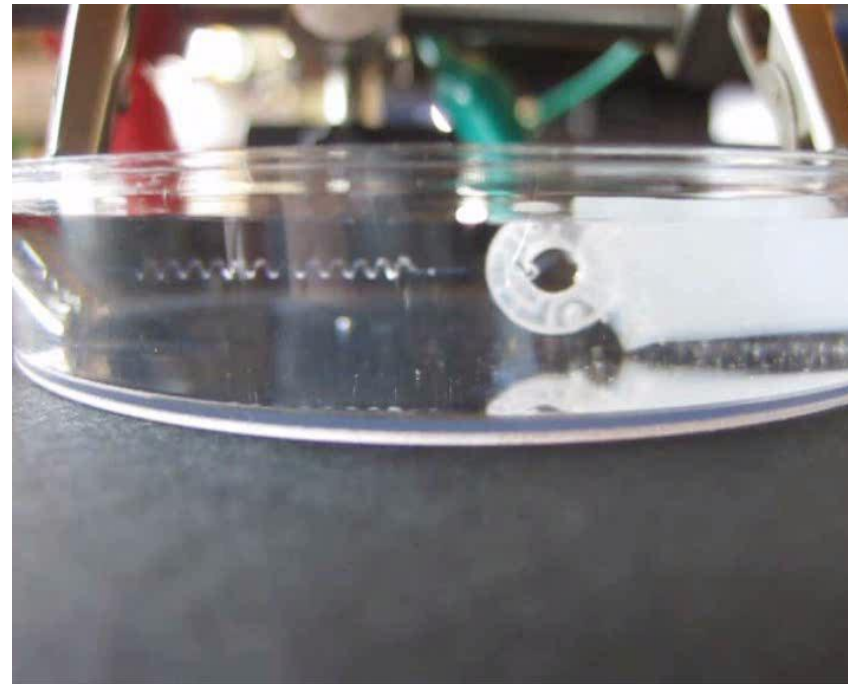


Case study

SMA actuator



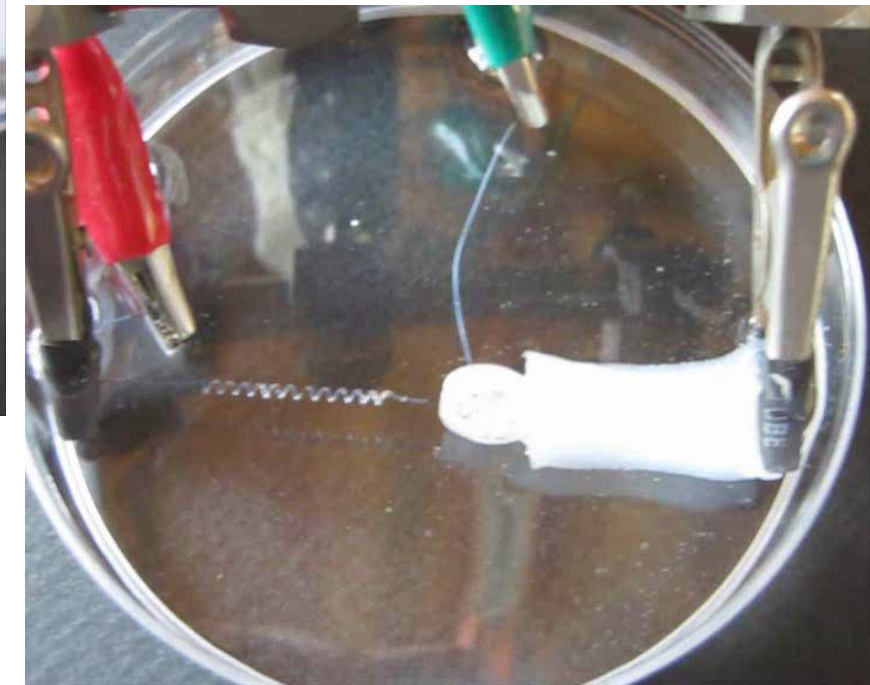
From a cool behaviour



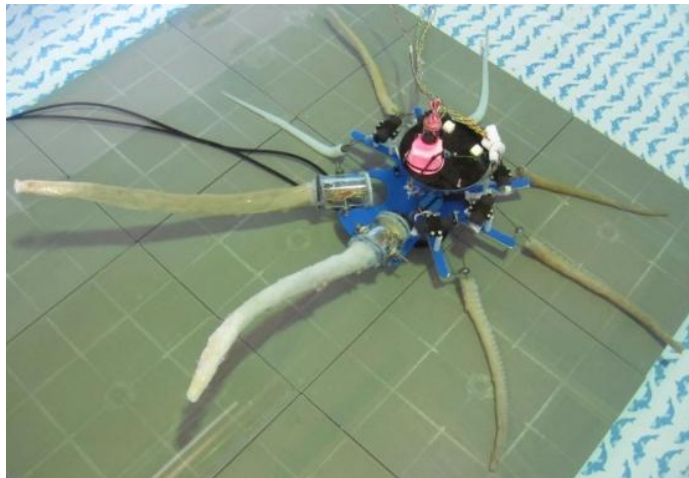
To an actuator
for soft robots

OCTOPUS IP (2009-2013)

Novel Design Principles and Technologies
for a New Generation of High Dexterity
Soft-bodied Robots Inspired by the
Morphology and Behaviour of the
Octopus



- About 1Hz of working frequency
- 1.2A of operating current
- PWM frequency: 100Hz
- PWM duty cycle: 25%
- $A_{\max}=2.4A$

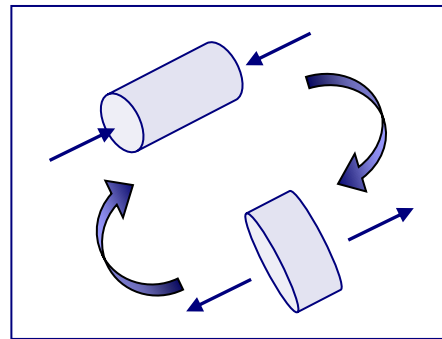


Octopus arm morphology

Longitudinal Muscles

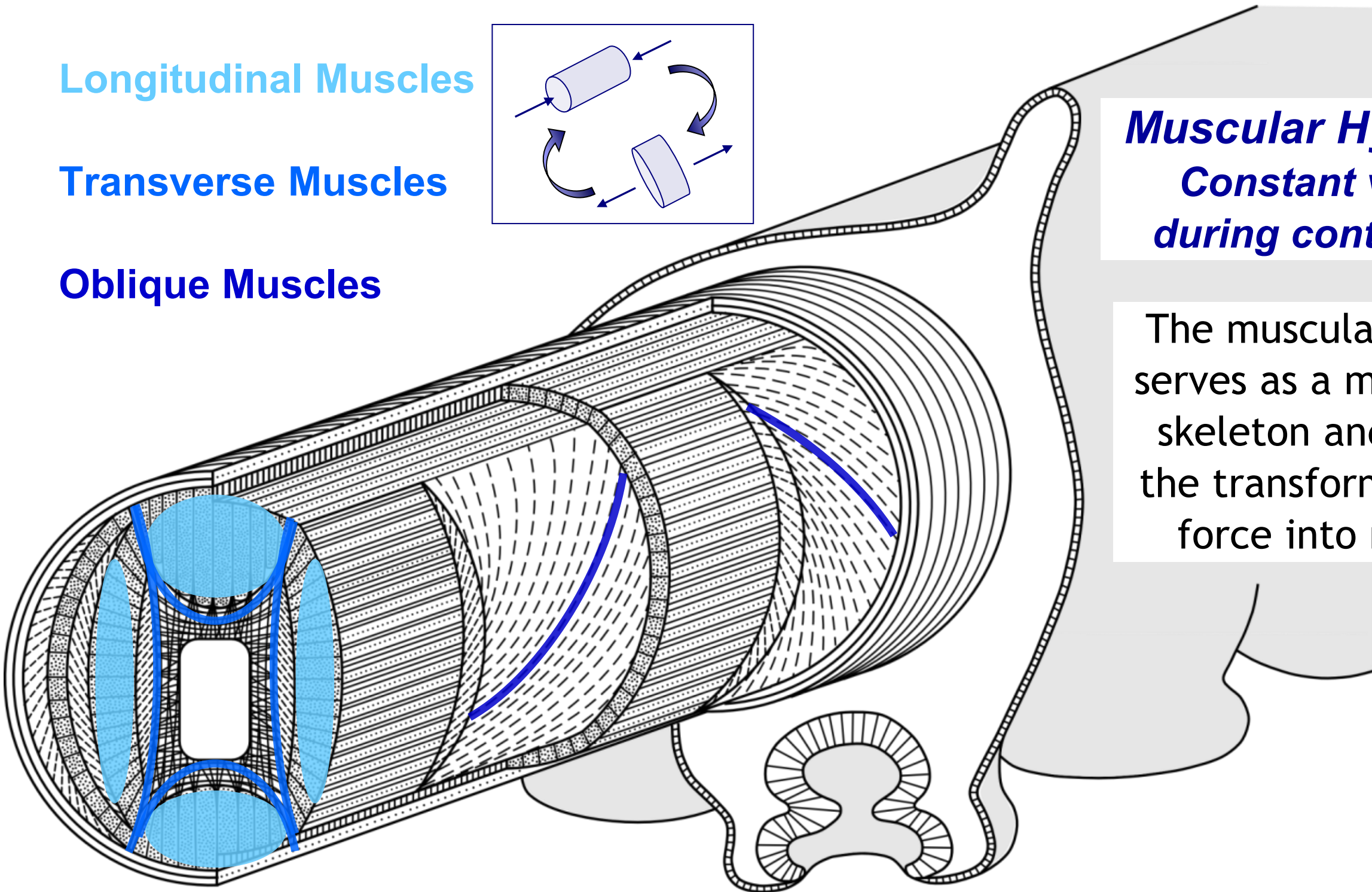
Transverse Muscles

Oblique Muscles



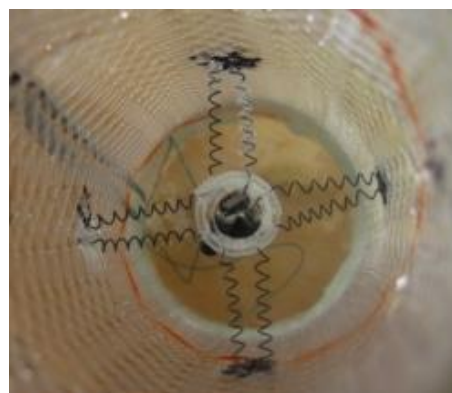
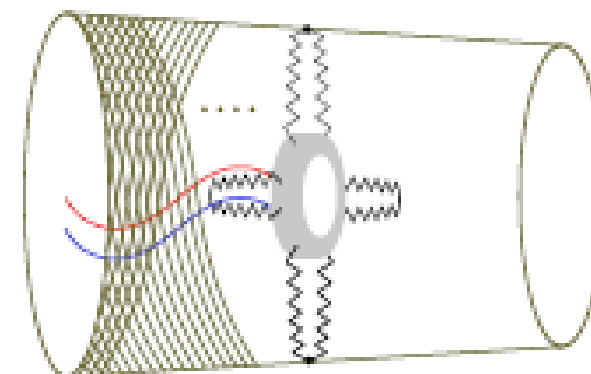
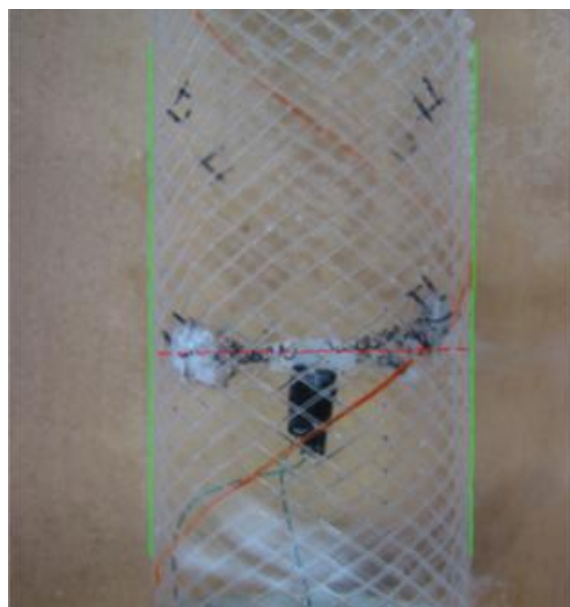
***Muscular Hydrostat:
Constant volume
during contractions***

The muscular system serves as a modifiable skeleton and allows the transformation of force into motion



The artificial muscular hydrostat

Modular unit

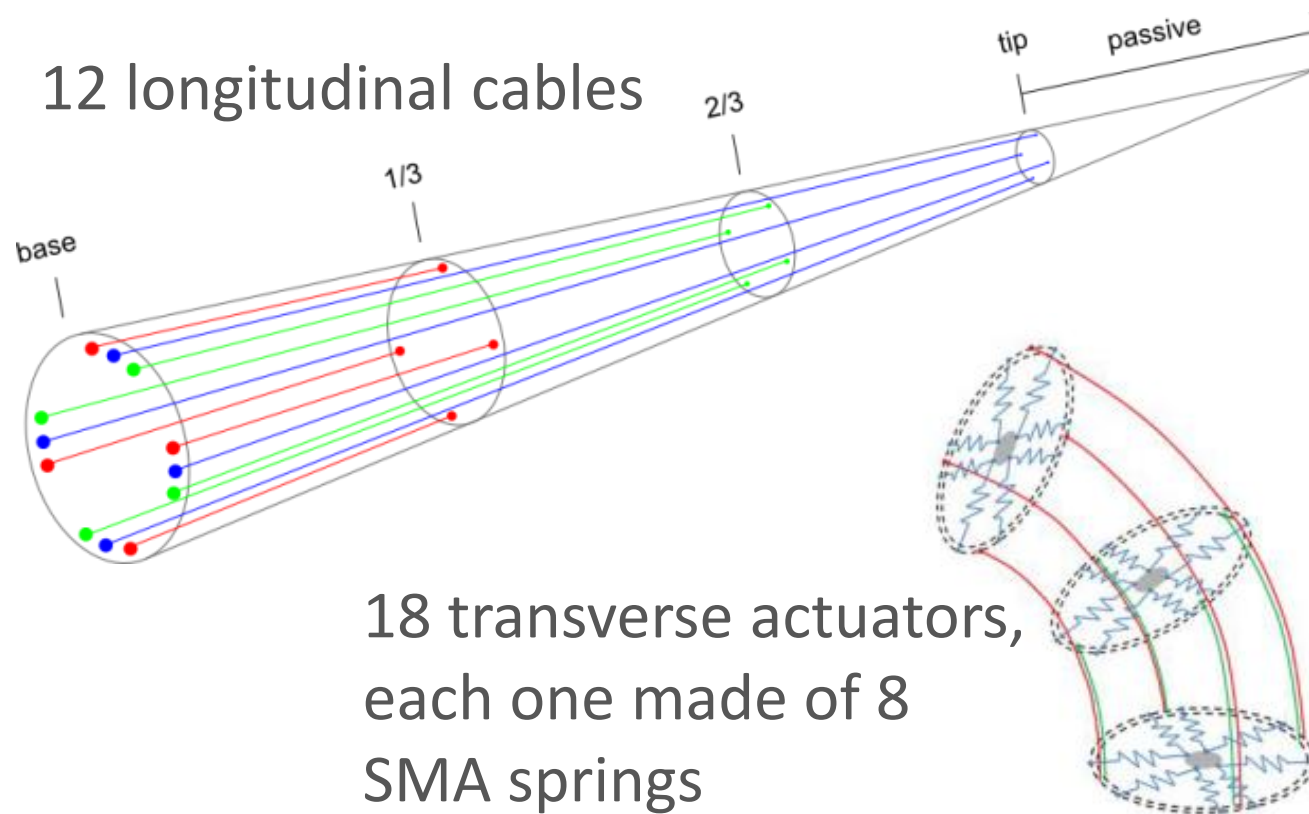


PET braided sleeve:
External diameter relaxed: 30mm
External diameter contracted: 20mm

Actuator:
8 SMA springs
0.2 mm SmartFlex® wire diameter
Spring internal diameter: 1 mm

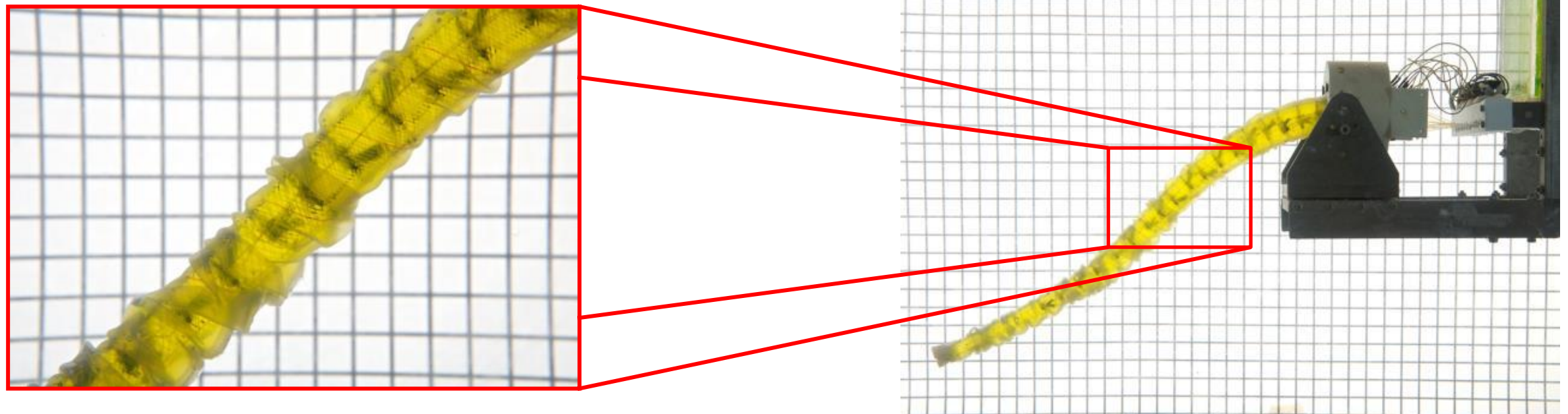


OCTOPUS arm v1

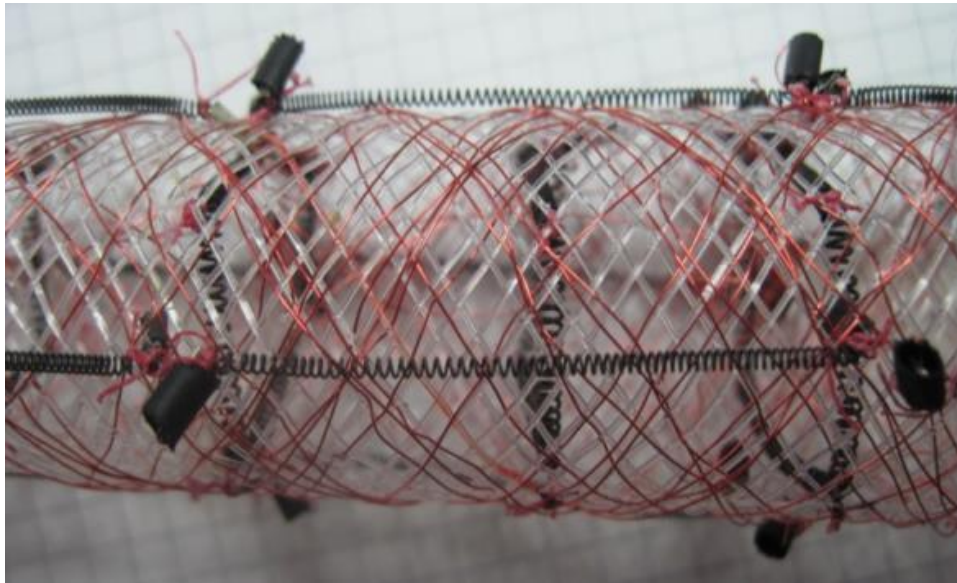


Design and development of a soft robotic octopus arm exploiting embodied intelligence

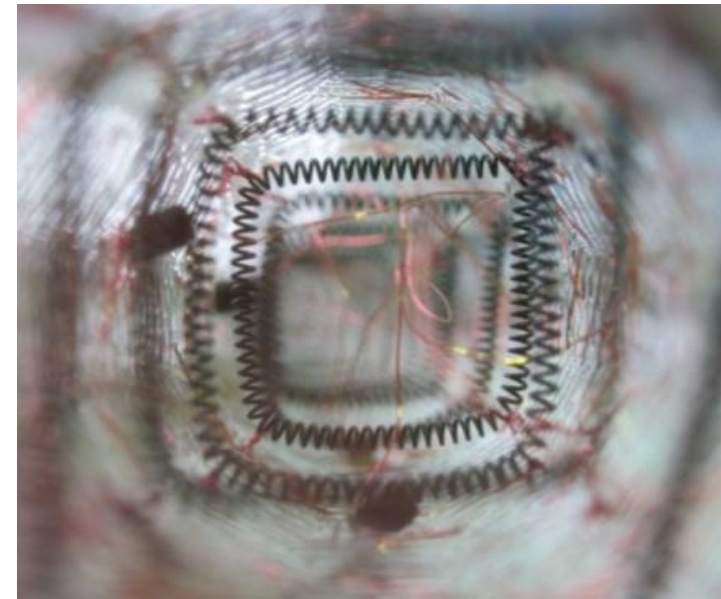
M. Cianchetti, M. Follador, B. Mazzolai, P. Dario, C. Laschi



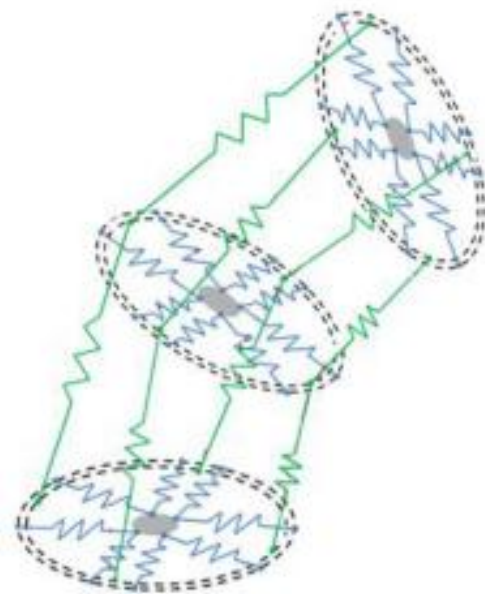
OCTOPUS arm v2



View of the external part of the arm



View of the internal part of the arm



Laschi et al., *Advanced Robotics*, 2012



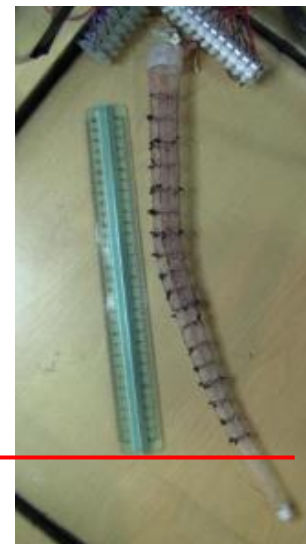
OCTOPUS arm v3



Rest length



Elongation



Shortening

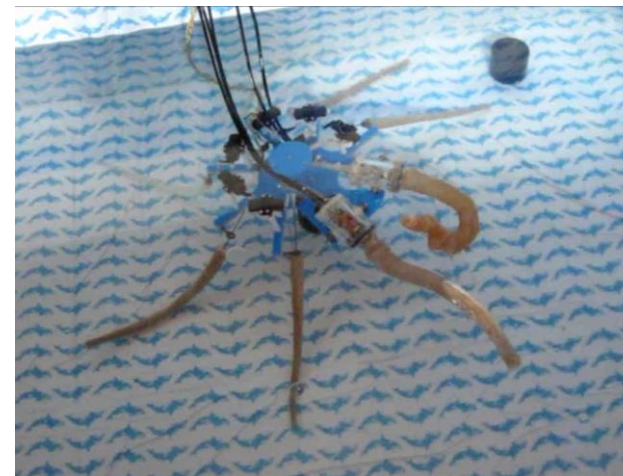
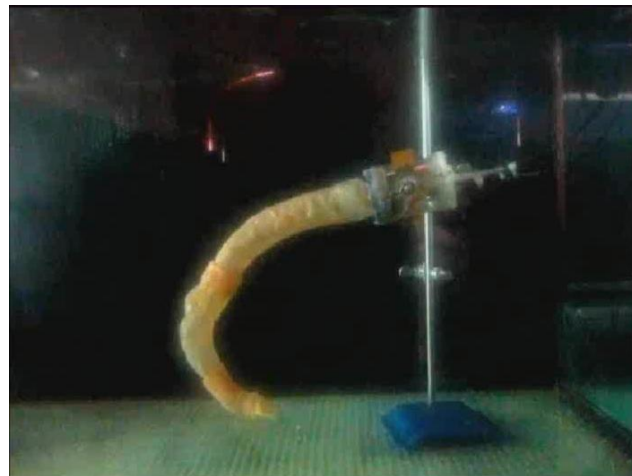
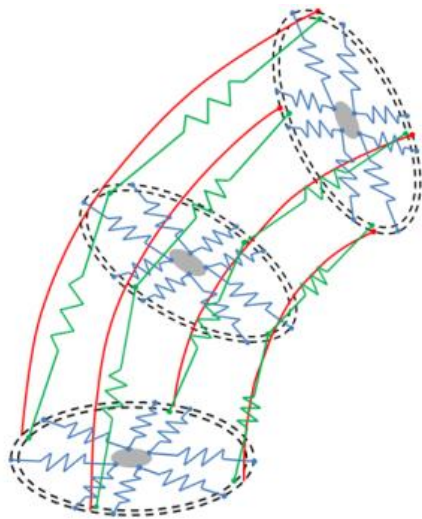


Bending



Multi – Bending

11 transverse and 12 longitudinal SMA springs, 4 longitudinal cables



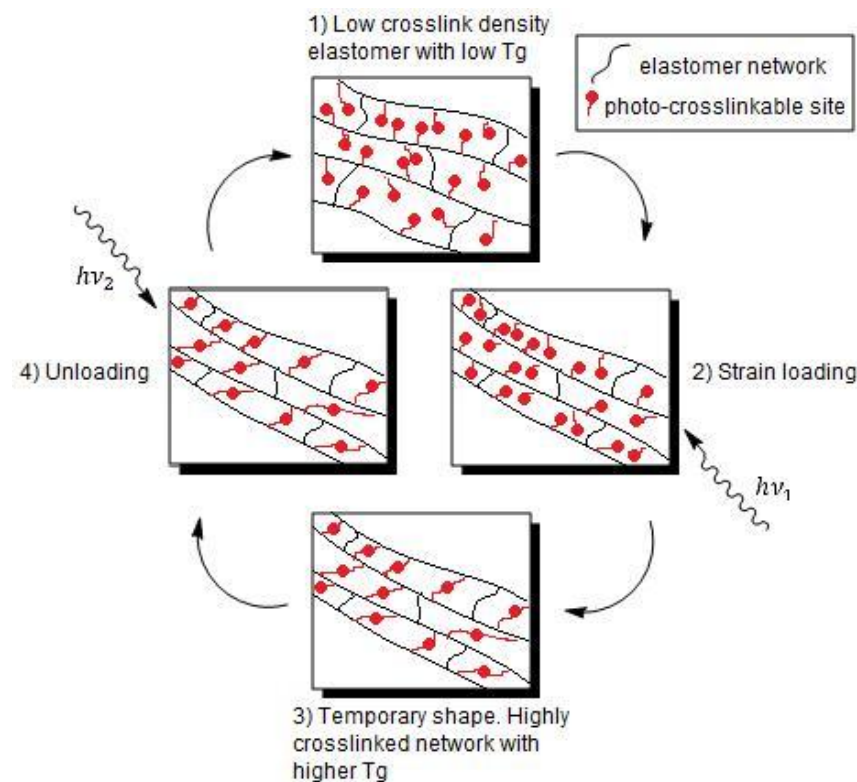
Cianchetti et al., *Bioinspiration & Biomimetics*, 2015



Shape Memory... Polymers



Shape Memory... Polymers



	SMPs	SMPs
Density (g/cm ³)	0.9–1.2	6–8
Extent of deformation	up to 800%	<8%
Required stress for deformation (MPa)	1–3	50–200
Stress generated upon recovery (MPa)	1–3	150–300
Transition temperatures (°C)	–10..100	–10..100
Recovery speed	1s – minutes	<1s
Processing conditions	<200 °C low pressure	>1000 °C high pressure
Costs	<\$10/lb	~\$250/lb



Shape-memory polymers differ from shape memory alloys by their **glass transition or melting transition** from a hard to a soft phase which is responsible for the shape-memory effect.

SMPs can retain two or sometimes three shapes, and the transition between those is induced by **temperature, electric or magnetic field, light or solution**.

